

Customer Churn in Telecommunications: A Statistical Analysis Using IBM SPSS Statistics

SPSS For Research (Final Project)

Shah Md. Tasrif Rahman

Introduction

Customer churn refers to the ending of an ongoing customer relationship with a service provider or business. For any business/service provider, churn can be examined using various information of the customer, their contract details, payment methods, and other related variables. This study uses a modified Telco Customer Churn dataset (Kleon, 2025) to examine patterns associated with whether a customer has left the company.

Research Questions and Hypotheses:

1. (Primary): Is customer tenure associated with customer churn?
 - H_0 : The distributions of customer tenure are the same for customers who were retained and customers who churned.
 - H_1 : The distributions of customer tenure differ between customers who were retained and customers who churned.
2. Are contract type and customer churn associated?
 - H_0 : Contract type and customer churn are independent.
 - H_1 : Contract type and customer churn are not independent, thus are associated.
3. Which customer characteristics are associated with the odds of customer churn?
 - H_0 : None of the selected customer characteristics is associated with the odds of customer churn, all corresponding regression coefficients are zero.
 - H_1 : At least one selected customer characteristic is associated with the odds of customer churn; at least one regression coefficient differs from zero.

Methods

Dataset and Sample

The dataset used in this study was the modified Telco Customer Churn Dataset obtained from Kaggle (Kleon, 2025). The raw CSV file contained 7043 customer records with 21 variables, including demographic characteristics, service variables, contract and payment information, continuous charge variables, and the binary churn outcome. Variable definitions and measurement levels are provided in Appendix A (Table A1).

This dataset was chosen because it meets all the criteria given for this project which are as follows:

- At least 6 usable variables, with a mix of categorical and continuous types.
- At least 150 cases (rows).
- Contains one binary/categorical outcome suitable for logistic regression, and at least one continuous variable suitable for a t-test/ANOVA and for correlation or linear regression.
- Publicly available and properly citable. Should not contain any personally identifying information such as phone number, name, etc.
- Some realistic data issues to clean (missing values, inconsistent coding, or duplicates) that will help demonstrate data cleaning skill.

Data Management and Cleaning

The raw CSV file was imported directly into SPSS and preserved as the source data. During the initial inspection, blank string values in gender, Partner, InternetService, and StreamingTV were not initially treated by SPSS as missing observations. These blank strings were therefore declared as user-missing values in SPSS. Numeric missing values already imported as system-missing were retained.

The dataset was checked for duplicate observations. No duplicate cases were identified, thus no records were removed for duplication.

All 7,043 observations were retained in the final dataset, individual analyses used the available cases required for their respective variables. Information about missing values is provided in Appendix A (Table A2).

The variable “Churn” was recoded into a binary analysis variable, with 0 representing retained customers and 1 representing churned customers. Categorical variables were defined with appropriate value labels and measurement levels.

Statistical Analysis

Descriptive statistics were produced for the available variables. Categorical variables were summarized using frequencies and percentages, whereas continuous variables were summarized using the mean, standard deviation, median, minimum, and maximum.

For the first or primary research question, customer tenure distributions were examined within churn groups using histograms, Q-Q plots, and normality tests. Because the distributions departed from normality, a Mann-Whitney U test was used to compare tenure between retained and churned customers.

For the second research question, a Pearson chi-square test of independence was used to examine the association between contract type and customer churn. Cramer's V was reported as effect-size measure, and expected cell counts were checked.

For the third research question, binary logistic regression was used with Churn as the dependent/target variable. Churn was recoded into a usable binary variable for this process. Predictors were selected based on their relevance to the target variable and their representation of different aspects of the customer relationship. The initial model included the predictors: tenure, MonthlyCharges, Contract, TechSupport, InternetService, and PaymentMethod. This model was specified with 11 expected predictor degrees of freedom. SPSS identified redundancy among the categorical predictors and reduced the estimable degrees of freedom to 10. Two reduced models were therefore examined to solve this issue, retaining either InternetService or TechSupport. The model retaining InternetService showed evidence of lack of fit according to the Hosmer-Lemeshow test, $\chi^2(8) = 17.656, p = .024$. The model retaining TechSupport showed no statistically significant evidence of lack of fit, $\chi^2(8) = 11.632, p = .168$. The latter was therefore selected as the final model. The final binary logistic regression model included the predictors: tenure, MonthlyCharges, PaymentMethod, Contract, and TechSupport.

Relevant graphs were produced accordingly to support the research questions.

Results

Table 1

Descriptive Statistics

Variable	Category	n	%	Mean (SD)	Median	Minimum	Maximum
Customer gender	Female	3,091	49.12				
	Male	3,202	50.88				
Indicates if the customer is a senior citizen (0 = No, 1 = Yes)	No	5,901	83.79				
	Yes	1,142	16.21				
Whether the customer has a partner	No	3,115	51.55				
	Yes	2,928	48.45				
Whether the customer has dependents	No	4,933	70.04				
	Yes	2,110	29.96				
Number of months the customer has stayed with the company				32.5 (24.5)	29.0	0.0	72.0
Whether the customer has phone service	No	682	9.68				
	Yes	6,361	90.32				

Variable	Category	n	%	Mean (SD)	Median	Minimum	Maximum
Whether the customer has multiple lines	No	3,390	48.13				
	No phone service	682	9.68				
	Yes	2,971	42.18				
Customer's internet service provider	DSL	2,082	34.45				
	Fiber optic	2,660	44.02				
	No	1,301	21.53				
Whether the customer has online security	No	3,498	49.67				
	No internet service	1,526	21.67				
	Yes	2,019	28.67				
Whether the customer has online backup	No	3,088	43.84				
	No internet service	1,526	21.67				
	Yes	2,429	34.49				
Whether the customer has device protection	No	3,095	43.94				
	No internet service	1,526	21.67				

Variable	Category	n	%	Mean (SD)	Median	Minimum	Maximum
Whether the customer has tech support	Yes	2,422	34.39				
	No	3,473	49.31				
	No internet service	1,526	21.67				
Whether the customer has streaming TV	Yes	2,044	29.02				
	No	2,199	39.67				
	No internet service	1,198	21.61				
Whether the customer has streaming movies	Yes	2,146	38.72				
	No	2,785	39.54				
	No internet service	1,526	21.67				
Type of contract	Yes	2,732	38.79				
	Month-to-month	3,875	55.02				
	One year	1,473	20.91				
Whether the customer uses paperless billing	Two year	1,695	24.07				
	No	2,872	40.78				

Variable	Category	n	%	Mean (SD)	Median	Minimum	Maximum
Payment method	Yes	4,171	59.22				
	Bank transfer (automatic)	1,544	21.92				
	Credit card (automatic)	1,522	21.61				
	Electronic check	2,365	33.58				
	Mailed check	1,612	22.89				
Monthly charges				64.88 (30.10)	70.55	18.25	118.60
Total charges to date				2283.30 (2266.77)	1397.48	18.80	8684.80
Whether the customer has left the company	Customer retained	5,174	73.46				
	Customer churned	1,869	26.54				

Note. N = 7,043. SD = standard deviation.

Table 2*Tests of Normality for Customer Tenure by Churn Status*

Customer Status	<i>n</i>	Kolmogorov-Smirnov <i>D</i>	<i>p</i>	Shapiro-Wilk <i>W</i>	<i>p</i>
Customer retained	3,352	.094	< .001	.917	< .001
Customer churned	1,191	.185	< .001	.831	< .001

Note. Dependent variable: Number of months the customer has stayed with the company. The Kolmogorov-Smirnov test uses the Lilliefors significance correction. For both retained and churned customers, the Kolmogorov-Smirnov and Shapiro-Wilk tests were statistically significant ($p < .001$), indicating deviations from normality.

Table 3*Mann-Whitney U Test Comparing Customer Tenure by Churn Status*

Churn status	<i>n</i>	Mean rank
Customer retained	3,352	2,546.27
Customer churned	1,191	1,500.08
Mann-Whitney <i>U</i>		1,076,759
<i>z</i>		-23.656
<i>p</i>		< .001

Note. The dependent variable was customer tenure (number of months the customer stayed with the company), and the grouping variable was churn status.

Table 4*Association Between Contract Type and Customer Churn Status*

Customer Status	Month-to-month	One year	Two year	Total
Customer retained n (%)	2,220 (57.3)	1,307 (88.7)	1,647 (97.2)	5,174 (73.5)
Customer churned n (%)	1,655 (42.7)	166 (11.3)	48 (2.8)	1,869 (26.5)
Total n (%)	3,875 (100.0)	1,473 (100.0)	1,695 (100.0)	7,043 (100.0)

Test Statistic	Value
Pearson χ^2	1184.597
<i>df</i>	2
<i>N</i>	7,043
<i>p</i>	< .001
Cramer's <i>V</i>	.410

Note. Customer churn status and contract type were significantly associated, $\chi^2(2, N = 7,043) = 1184.597, p < .001$, and Cramer's *V* = .410.

Table 5*Binary Logistic Regression Model Statistics*

Statistic	Value
Omnibus χ^2 (df)	1291.125 (9), $p < .001$
-2 Log Likelihood	3936.103
Cox & Snell R^2	.247
Nagelkerke R^2	.362
Hosmer-Lemeshow χ^2 (df)	11.632 (8), $p = .168$
Classification Accuracy	79.2%

Note. χ^2 = chi-square test statistic. Cox & Snell and Nagelkerke values represent pseudo- R^2 measures. A non-significant Hosmer-Lemeshow test indicates adequate model fit. Classification accuracy represents the percentage of cases correctly classified using a cutoff value of .500. Classification performance was higher for customers who did not churn (89.6%) than for customers who churned (49.9%).

Table 6*Regression Coefficients*

Predictor	<i>B</i>	SE	Wald χ^2	df	<i>p</i>	OR	95% CI for OR
Number of months the customer has stayed with the company	-0.029	0.003	133.258	1	< .001	0.971	[0.966, 0.976]
Monthly charges	0.022	0.002	95.177	1	< .001	1.022	[1.017, 1.026]
Payment method (Reference: Bank transfer [automatic])			32.633	3	< .001		
Credit card (automatic)	-0.049	0.137	0.127	1	.721	0.952	[0.728, 1.246]
Electronic check	0.412	0.112	13.483	1	< .001	1.510	[1.212, 1.881]
Mailed check	-0.120	0.136	0.781	1	.377	0.887	[0.679, 1.158]
Type of contract (Reference: Month-to-month)			95.580	2	< .001		
One year	-0.869	0.124	49.299	1	< .001	0.419	[0.329, 0.534]
Two year	-1.797	0.212	71.909	1	< .001	0.166	[0.109, 0.251]
Whether the customer has tech support (Reference: No)			26.076	2	< .001		
No internet service	-0.466	0.174	7.142	1	.008	0.628	[0.446, 0.883]
Yes	-0.470	0.102	21.078	1	< .001	0.625	[0.511, 0.764]
Constant	-1.277	0.184	47.933	1	< .001	0.279	

Note. B = logistic regression model coefficient; SE = standard error; OR = odds ratio (Exp(B)); CI = confidence interval. Reference categories were Bank transfer (automatic) for payment method, Month-to-month for contract type, and No for tech support.

Table 7*Binary Logistic Regression Model Selection*

Model	Modification	H-L test χ^2	df	<i>p</i>	Model Status
Initial model	None	5.454	8	.708	Redundancy detected in predictors
InternetService retained	TechSupport removed	17.656	8	.024	Evidence of lack of fit
TechSupport retained	InternetService removed	11.632	8	.168	Final model retained

Note. Model selection was based on redundancy diagnostics and Hosmer-Lemeshow goodness-of-fit tests. The final model retained the variables: tenure, MonthlyCharges, PaymentMethod, Contract, and TechSupport. A non-significant Hosmer-Lemeshow test indicated adequate model fit.

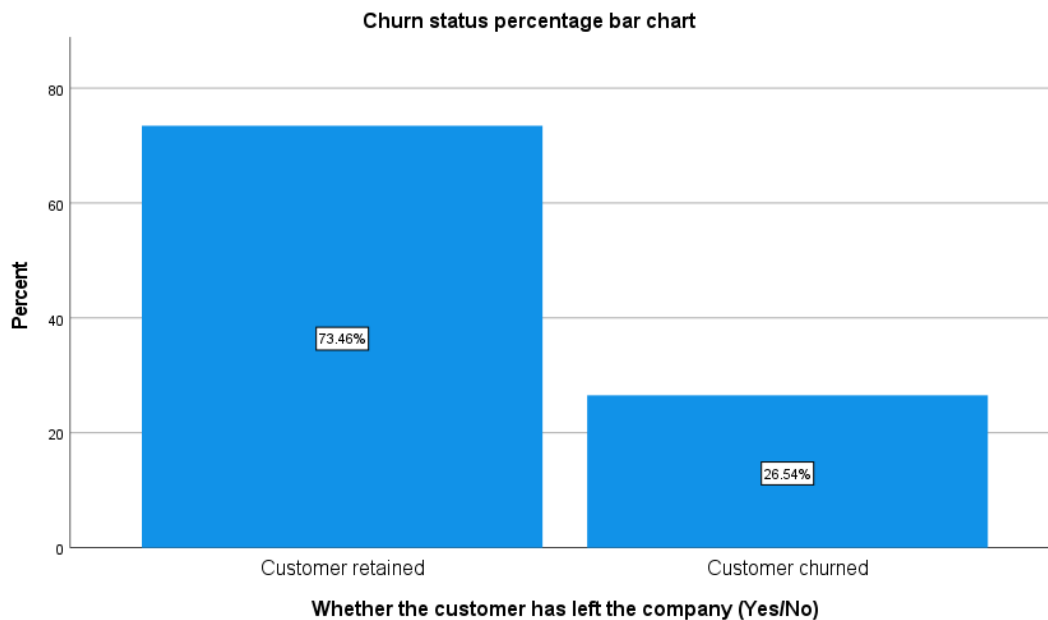
Figure 1*Distribution of customer churn status*

Figure 2

Distribution of customer tenure by split by churn status

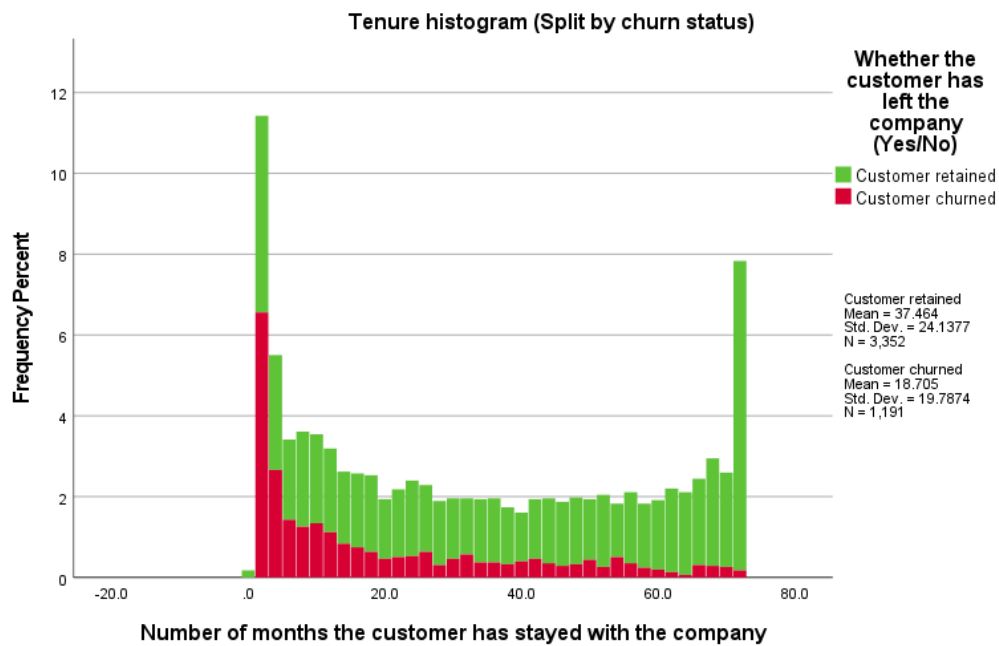


Figure 3

Customer tenure boxplot by churn status

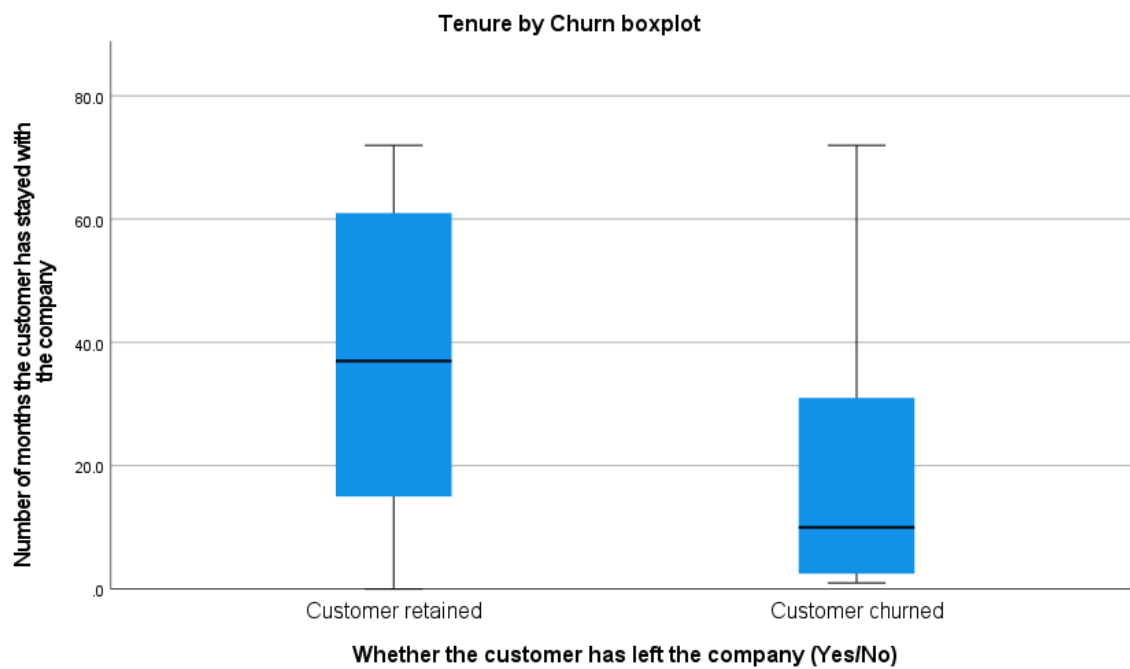


Figure 4

Churn status by contract type, expressed as percentages within each contract category

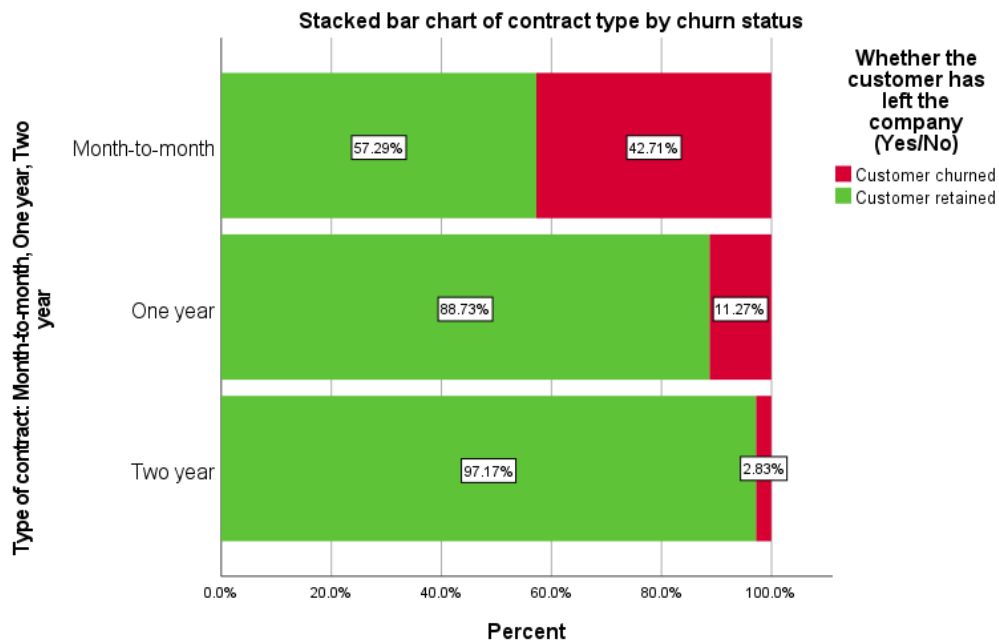


Figure 5

Churn status by payment method, expressed as percentages within each payment category

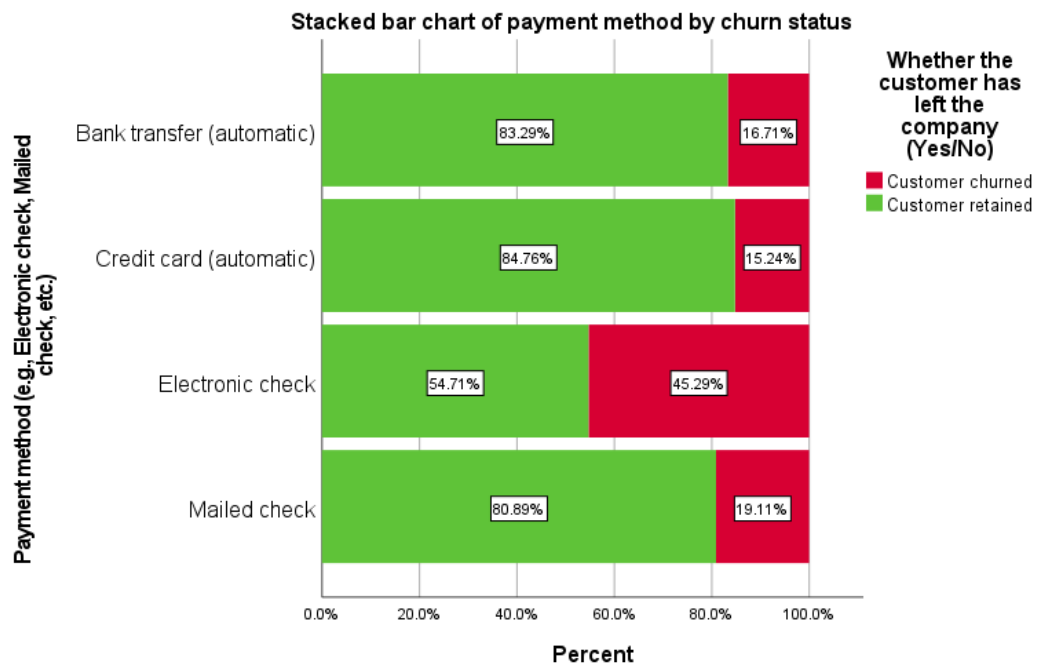
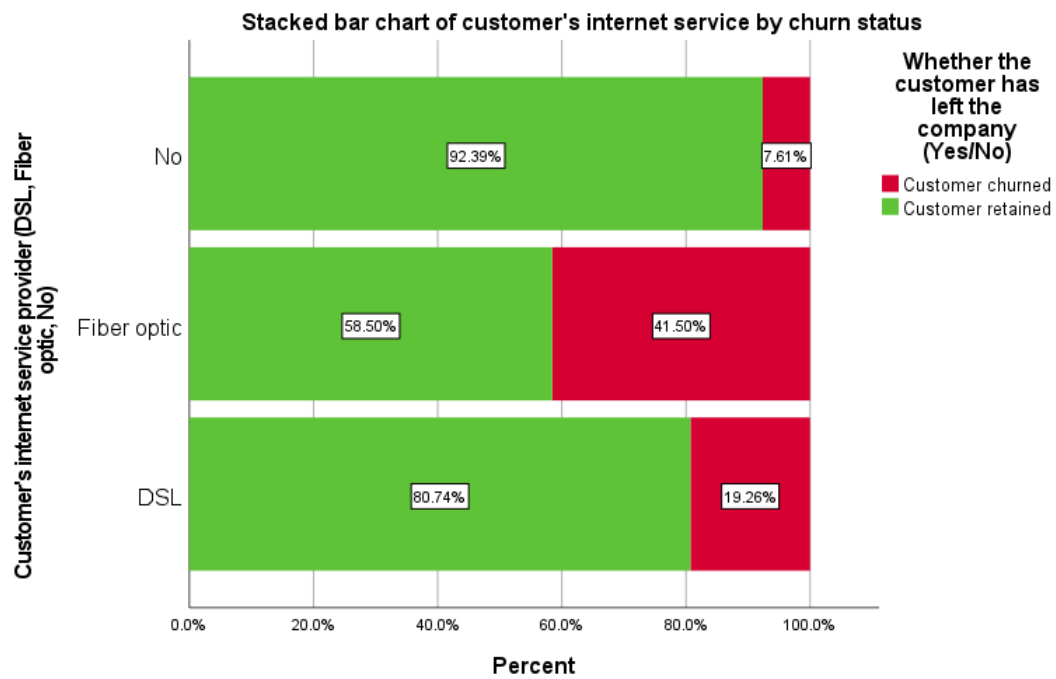


Figure 6

Churn status by internet service type, expressed as percentages within each service category



Discussion

This study examined customer churn in a telecommunications company dataset using descriptive statistics, non-parametric testing, chi-square testing, and binary logistic regression. Three research questions were used to investigate whether customer tenure and contract type were associated with customer churn and whether selected customer characteristics were associated with the odds of customer churn.

The first research question examined the association between customer tenure and churn. The Mann-Whitney U test showed a statistically significant difference in tenure distributions between retained and churned customers. This finding suggests that churned customers had substantially shorter tenure distributions than retained customers.

The second research question examined the relationship between contract type and churn. A significant association was found between the two variables, $\chi^2(2, N = 7,043) = 1184.597, p < .001$, Cramer's $V = .410$. The observed churn rates differed considerably across contract categories: 42.7% among month-to-month customers, 11.3% among one-year customers, and 2.8% among two-year customers. These results indicate that churn status varied substantially across contract types. However, because this analysis is observational, these results indicate an association rather than a causal effect of contract type on churn.

The third research question examined several customer characteristics associated with the odds of churn using binary logistic regression. The final model included tenure, MonthlyCharges, PaymentMethod, Contract, and TechSupport and was statistically significant, $\chi^2(9) = 1291.125, p < .001$. The model had a Nagelkerke pseudo- R^2 of .362, while the Hosmer-Lemeshow test was non-significant, $\chi^2(8) = 11.632, p = .168$, providing no evidence of lack of fit. Tenure, MonthlyCharges, contract type, TechSupport, and one PaymentMethod (Electronic check) showed statistically significant associations with odds of customer churn. In particular, longer tenure was associated with lower odds of churn, whereas higher MonthlyCharges were associated with higher odds of churn. Compared with month-to-month contracts, both one-year and two-year contracts were associated with lower estimated odds of churn. And compared to Bank transfer (automatic), only the Electronic check category was associated with significantly higher estimated odds of churn.

The findings from these three research questions are broadly consistent with one another. The descriptive statistics and inferential results indicate that customer relationship duration and contractual characteristics are closely related to churn status, while the logistic regression shows that several characteristics remain associated with churn when considered together.

Some limitations of this study are as follows:

1. The regression analysis included only 4,543 of the 7,043 observations because of missing tenure values, meaning that 35.5% of the records were not included in the multivariable regression analysis, which may affect the representativeness of the regression estimates.

2. The dataset contains intentionally introduced missing values by the dataset author, so its data structure may not represent naturally occurring customer data.
3. Although the binary logistic regression model correctly classified 79.2% of cases overall, it correctly identified retained customers more often (89.6%) than churned customers (49.9%). Therefore, overall accuracy alone does not represent performance equally across the two outcome groups.

Overall, the analysis provides evidence that customer tenure, contract type, services, billing, and payment characteristics are associated with customer churn in this dataset.

References

Kleon, A. (2025, July 18).  *Telco Customer Churn Dataset*. Kaggle. Telco Customer Churn Dataset. <https://www.kaggle.com/datasets/jethwaaatmik/telco-customer-churn-dataset>

Appendix A

Additional Tables

Table A1

Dataset Variables and Measurement Levels

Variable	Description	Unit	Measure
customerID	Unique identifier for each customer		Nominal
gender	Customer gender: Male/Female		Nominal
SeniorCitizen	Indicates if the customer is a senior citizen (0 = No, 1 = Yes)		Nominal
Partner	Whether the customer has a partner		Nominal
Dependents	Whether the customer has dependents		Nominal
tenure	Number of months the customer has stayed with the company	Months	Scale
PhoneService	Whether the customer has phone service		Nominal
MultipleLines	Whether the customer has multiple lines		Nominal
InternetService	Customer's internet service provider (DSL, Fiber optic, No)		Nominal
OnlineSecurity	Whether the customer has online security		Nominal
OnlineBackup	Whether the customer has online backup		Nominal
DeviceProtection	Whether the customer has device protection		Nominal
TechSupport	Whether the customer has tech support		Nominal
StreamingTV	Whether the customer has streaming TV		Nominal
StreamingMovies	Whether the customer has streaming movies		Nominal
Contract	Type of contract: Month-to-month, One year, Two year		Nominal
PaperlessBilling	Whether the customer uses paperless billing		Nominal
PaymentMethod	Payment method (e.g., Electronic check, Mailed check, etc.)		Nominal
MonthlyCharges	Monthly charges	USD	Scale
TotalCharges	Total charges to date	USD	Scale
Churn	Whether the customer has left the company (Yes/No)		Nominal

Table A2*Missing Data Summary*

Variable	Missing (n)	Missing (%)
gender	750	10.65
Partner	1,000	14.20
tenure	2,500	35.50
InternetService	1,000	14.20
StreamingTV	1,500	21.30
MonthlyCharges	1,500	21.30
TotalCharges	11	0.16

Note. Missing (%) represents the percentage of observations with missing data for each variable.

Appendix B

SPSS Output

* Encoding: UTF-8.

* /Step-1: Data loading/

PRESERVE.

SET DECIMAL DOT.

GET DATA /TYPE=TXT

/FILE="E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final
Project\telco_data.csv"

/ENCODING='UTF8'

/DELIMITERS=","

/QUALIFIER="'"

/ARRANGEMENT=DELIMITED

/FIRSTCASE=2

/DATATYPEMIN PERCENTAGE=95.0

/VARIABLES=

customerID AUTO

gender AUTO

SeniorCitizen AUTO

Partner AUTO

Dependents AUTO

tenure AUTO

PhoneService AUTO

MultipleLines AUTO
InternetService AUTO
OnlineSecurity AUTO
OnlineBackup AUTO
DeviceProtection AUTO
TechSupport AUTO
StreamingTV AUTO
StreamingMovies AUTO
Contract AUTO
PaperlessBilling AUTO
PaymentMethod AUTO
MonthlyCharges AUTO
TotalCharges AUTO
Churn AUTO

/MAP.

RESTORE.

CACHE.

EXECUTE.

Data written to the working file.

21 variables and 7043 cases written.

Variable: customerID Type: String Format : A10

Variable: gender Type: String Format : A6

Variable: SeniorCitizen Type: Number Format : F1

Variable: Partner Type: String Format : A3

Variable: Dependents Type: String Format : A3

Variable: tenure Type: Number Format : F4.1 One or more values were set to system-missing.

Variable: PhoneService Type: String Format : A3

Variable: MultipleLines Type: String Format : A16

Variable: InternetService Type: String Format : A11

Variable: OnlineSecurity Type: String Format : A19

Variable: OnlineBackup Type: String Format : A19

Variable: DeviceProtection Type: String Format : A19

Variable: TechSupport Type: String Format : A19

Variable: StreamingTV Type: String Format : A19

Variable: StreamingMovies Type: String Format : A19

Variable: Contract Type: String Format : A14

Variable: PaperlessBilling Type: String Format : A3

Variable: PaymentMethod Type: String Format : A25

Variable: MonthlyCharges Type: Number Format : F6.2 One or more values were set to system-missing.

Variable: TotalCharges Type: Number Format : F7.2 One or more values were set to system-missing.

Variable: Churn Type: String Format : A3

Substitute the following to build syntax for these data.

/VARIABLES=

customerID A10

gender A6

SeniorCitizen F1

Partner A3

Dependents A3

tenure F4.1

PhoneService A3

MultipleLines A16

InternetService A11

OnlineSecurity A19

OnlineBackup A19

DeviceProtection A19

TechSupport A19

StreamingTV A19

StreamingMovies A19

Contract A14

PaperlessBilling A3

PaymentMethod A25

MonthlyCharges F6.2

TotalCharges F7.2

Churn A3

DATASET NAME DataSet1 WINDOW=FRONT.

Dataset Name

Warnings

The active dataset will replace the existing dataset named DataSet1.

SAVE OUTFILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final Project\Final_Project.sav'

/COMPRESSED.

* /Step-2: Initial data Checking/

DATASET ACTIVATE DataSet1.

FREQUENCIES VARIABLES=customerID gender SeniorCitizen Partner Dependents tenure
PhoneService

MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection
TechSupport StreamingTV

StreamingMovies Contract PaperlessBilling PaymentMethod MonthlyCharges
TotalCharges Churn

/FORMAT=NOTABLE

/STATISTICS=RANGE MINIMUM MAXIMUM

/ORDER=ANALYSIS.

Frequencies

		Statistics																				
		customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingMovies	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
N	Valid	7040	7040	7040	7040	7040	4563	7040	7040	7040	7040	7040	7040	7040	7040	7040	7040	7040	7040	5543	7032	7040
	Missing	0	0	0	0	0	280	0	0	0	0	0	0	0	0	0	0	0	0	190	11	0
Range				1			72.0													100.35	8006.00	
Minimum				0			.0													16.25	16.00	
Maximum				1			72.0													115.00	8006.00	

* /Step-3: Fixing "SPSS failing to find missing value properly" issue/

MISSING VALUES gender Partner InternetService StreamingTV (' ').

* Checking again if things are okay or not

DATASET ACTIVATE DataSet1.

FREQUENCIES VARIABLES=customerID gender SeniorCitizen Partner Dependents tenure
PhoneService

MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection
TechSupport StreamingTV

StreamingMovies Contract PaperlessBilling PaymentMethod MonthlyCharges
TotalCharges Churn

/FORMAT=NOTABLE

/STATISTICS=RANGE MINIMUM MAXIMUM

/ORDER=ANALYSIS.

Frequencies

		Statistics																					
		customerID	gender	SeniorCitizen	Partner	Dependents	Worse	PhoneService	MultipleLines	InternetService	OnlineSecurity	OnlineBackup	DeviceProtection	TechSupport	StreamingTV	StreamingBox	x	Contract	PaperlessBilling	PaymentMethod	MonthlyCharges	TotalCharges	Churn
N	Valid	7043	6283	7043	6543	7543	4963	7043	7543	6543	7043	7043	7043	7043	5543	7043	7043	7043	7043	7043	5543	7032	7043
	Missing	0	760	0	1000	0	2500	0	0	1000	0	0	0	0	1000	0	0	0	0	0	1500	11	0
Range				1			72.0														93.35	8856.50	
Minimum				0			.0														18.25	18.80	
Maximum				1			72.0														118.80	8856.50	

* /Step-4: Checking or fixing duplicate observations/

DATASET ACTIVATE DataSet1.

* Identify Duplicate Cases.

SORT CASES BY customerID(A).

MATCH FILES

/FILE=*

```

/BY customerID
/FIRST=PrimaryFirst
/LAST=PrimaryLast.
DO IF (PrimaryFirst).
COMPUTE MatchSequence=1-PrimaryLast.
ELSE.
COMPUTE MatchSequence=MatchSequence+1.
END IF.
LEAVE MatchSequence.
FORMATS MatchSequence (f7).
COMPUTE InDupGrp=MatchSequence>0.
SORT CASES InDupGrp(D).
MATCH FILES
/FILE=*
/DROP=PrimaryFirst InDupGrp MatchSequence.
VARIABLE LABELS PrimaryLast 'Indicator of each last matching case as Primary'.
VALUE LABELS PrimaryLast 0 'Duplicate Case' 1 'Primary Case'.
VARIABLE LEVEL PrimaryLast (ORDINAL).
FREQUENCIES VARIABLES=PrimaryLast.

```

Frequencies

Statistics

Indicator of each last
matching case as
Primary

N	Valid	7043
	Missing	0

Indicator of each last matching case as Primary

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Primary Case	7043	100.0	100.0	100.0

EXECUTE.

* No duplicates exist, splendid.

* /Step-4: Data labelling/

VARIABLE LABELS

customerID "Unique identifier for each customer"

gender "Customer gender: Male/Female"

SeniorCitizen "Indicates if the customer is a senior citizen (0 = No, 1 = Yes)"

Partner "Whether the customer has a partner"

Dependents "Whether the customer has dependents"

tenure "Number of months the customer has stayed with the company"

PhoneService "Whether the customer has phone service"

MultipleLines "Whether the customer has multiple lines"

InternetService "Customer's internet service provider (DSL, Fiber optic, No)"

OnlineSecurity "Whether the customer has online security"

OnlineBackup "Whether the customer has online backup"

DeviceProtection "Whether the customer has device protection"

TechSupport "Whether the customer has tech support"

StreamingTV "Whether the customer has streaming TV"

StreamingMovies "Whether the customer has streaming movies"

Contract "Type of contract: Month-to-month, One year, Two year"

PaperlessBilling "Whether the customer uses paperless billing"

PaymentMethod "Payment method (e.g., Electronic check, Mailed check, etc.)"

MonthlyCharges "Monthly charges"

TotalCharges "Total charges to date"

Churn "Whether the customer has left the company (Yes/No)".

EXECUTE.

VALUE LABELS

SeniorCitizen

0 "No"

1 "Yes".

VALUE LABELS

Churn

"No" "Customer retained"

"Yes" "Customer churned".

EXECUTE.

* Things are okay now, we can move onto any required further processing and analysis.

SAVE OUTFILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final
Project\Final_Project.sav'

/COMPRESSED.

* /Task-1: Table-1/

* Custom Tables.

CTABLES

/VLABELS VARIABLES=gender SeniorCitizen Partner Dependents tenure PhoneService
MultipleLines

InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport
StreamingTV

StreamingMovies Contract PaperlessBilling PaymentMethod MonthlyCharges
TotalCharges Churn

DISPLAY=LABEL

/TABLE gender [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] +
SeniorCitizen

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + Partner [COUNT
'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + Dependents

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + tenure [MEAN,
STDDEV, MEDIAN,

MINIMUM, MAXIMUM] + PhoneService [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] +

MultipleLines [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + InternetService

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + OnlineSecurity [COUNT 'Frequency'

F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + OnlineBackup [COUNT 'Frequency' F40.0, TABLEPCT.COUNT

'Percentage' PCT40.2] + DeviceProtection [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage'

PCT40.2] + TechSupport [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + StreamingTV

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + StreamingMovies [COUNT 'Frequency'

F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + Contract [COUNT 'Frequency' F40.0, TABLEPCT.COUNT

'Percentage' PCT40.2] + PaperlessBilling [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage'

PCT40.2] + PaymentMethod [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] +

MonthlyCharges [MEAN, STDDEV, MEDIAN, MINIMUM, MAXIMUM] + TotalCharges [MEAN, STDDEV, MEDIAN,

MINIMUM, MAXIMUM] + Churn [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2]

/CATEGORIES VARIABLES=gender Partner Dependents PhoneService MultipleLines InternetService

OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies Contract

PaperlessBilling PaymentMethod ORDER=A KEY=VALUE EMPTY=EXCLUDE

/CATEGORIES VARIABLES=SeniorCitizen Churn ORDER=A KEY=VALUE EMPTY=INCLUDE

/CRITERIA CILEVEL=95.

Custom Tables

		Frequency	Percentage	Mean	Standard Deviation	Median	Minimum	Maximum
Customer gender: Male/Female	Female	3091	49.12%					
	Male	3202	50.88%					
Indicates if the customer is a senior citizen (0 = No, 1 = Yes)	No	5901	83.79%					
	Yes	1142	16.21%					
Whether the customer has a partner	No	3115	51.55%					
	Yes	2928	48.45%					
Whether the customer has dependents	No	4933	70.04%					
	Yes	2110	29.96%					
Number of months the customer has stayed with the company				32.5	24.5	29.0	.0	72.0
Whether the customer has phone service	No	682	9.68%					
	Yes	6361	90.32%					
Whether the customer has multiple lines	No	3390	48.13%					
	No phone service	682	9.68%					
	Yes	2971	42.18%					

Customer's internet service provider (DSL, Fiber optic, No)	DSL	2082	34.45%					
	Fiber optic	2660	44.02%					
	No	1301	21.53%					
Whether the customer has online security	No	3498	49.67%					
	No internet service	1526	21.67%					
	Yes	2019	28.67%					
Whether the customer has online backup	No	3088	43.84%					
	No internet service	1526	21.67%					
	Yes	2429	34.49%					
Whether the customer has device protection	No	3095	43.94%					
	No internet service	1526	21.67%					
	Yes	2422	34.39%					
Whether the customer has tech support	No	3473	49.31%					
	No internet service	1526	21.67%					
	Yes	2044	29.02%					
Whether the customer has streaming TV	No	2199	39.67%					
	No internet service	1198	21.61%					
	Yes	2146	38.72%					
Whether the customer has streaming movies	No	2785	39.54%					
	No internet service	1526	21.67%					
	Yes	2732	38.79%					
Type of contract: Month-to-month, One year, Two year	Month-to-month	3875	55.02%					
	One year	1473	20.91%					
	Two year	1695	24.07%					
	No	2872	40.78%					

Whether the customer uses paperless billing	Yes	4171	59.22%					
Payment method (e.g., Electronic check, Mailed check, etc.)	Bank transfer (automatic)	1544	21.92%					
	Credit card (automatic)	1522	21.61%					
	Electronic check	2365	33.58%					
	Mailed check	1612	22.89%					
Monthly charges				64.88	30.10	70.55	18.25	118.60
Total charges to date				2283.30	2266.77	1397.48	18.80	8684.80
Whether the customer has left the company (Yes/No)	Customer retained	5174	73.46%					
	Customer churned	1869	26.54%					

* /Task-2: Testing research questions/

* /RQ-1: Is customer tenure associated with customer churn?/

* Testing normality

EXAMINE VARIABLES=tenure BY Churn

/PLOT BOXPLOT HISTOGRAM NPLOT

/COMPARE GROUPS

/STATISTICS DESCRIPTIVES

/CINTERVAL 95

/MISSING LISTWISE

/NOTOTAL.

Explore

Whether the customer has left the company (Yes/No)

Case Processing Summary

		Cases					
		Valid		Missing		Total	
Whether the customer has left the company (Yes/No)		N	Percent	N	Percent	N	Percent
Number of months the customer has stayed with the company	Customer retained	3352	64.8%	1822	35.2%	5174	100.0%
	Customer churned	1191	63.7%	678	36.3%	1869	100.0%

Descriptives

Whether the customer has left the company (Yes/No)			Statistic	Std. Error
Number of months the customer has stayed with the company	Customer retained	Mean	37.464	.4169
		95% Confidence Interval for Mean		
		Lower Bound		36.647
		Upper Bound		38.282

	5% Trimmed Mean		37.570	
	Median		37.000	
	Variance		582.628	
	Std. Deviation		24.1377	
	Minimum		.0	
	Maximum		72.0	
	Range		72.0	
	Interquartile Range		46.0	
	Skewness		-.021	.042
	Kurtosis		-1.419	.085
	Customer churned	Mean		18.705
95% Confidence Interval for Mean		Lower Bound	17.580	
		Upper Bound	19.830	
5% Trimmed Mean		17.020		
Median		10.000		
Variance		391.541		
Std. Deviation		19.7874		
Minimum		1.0		
Maximum		72.0		
Range		71.0		
Interquartile Range		29.0		
Skewness		1.059	.071	
Kurtosis		-.053	.142	

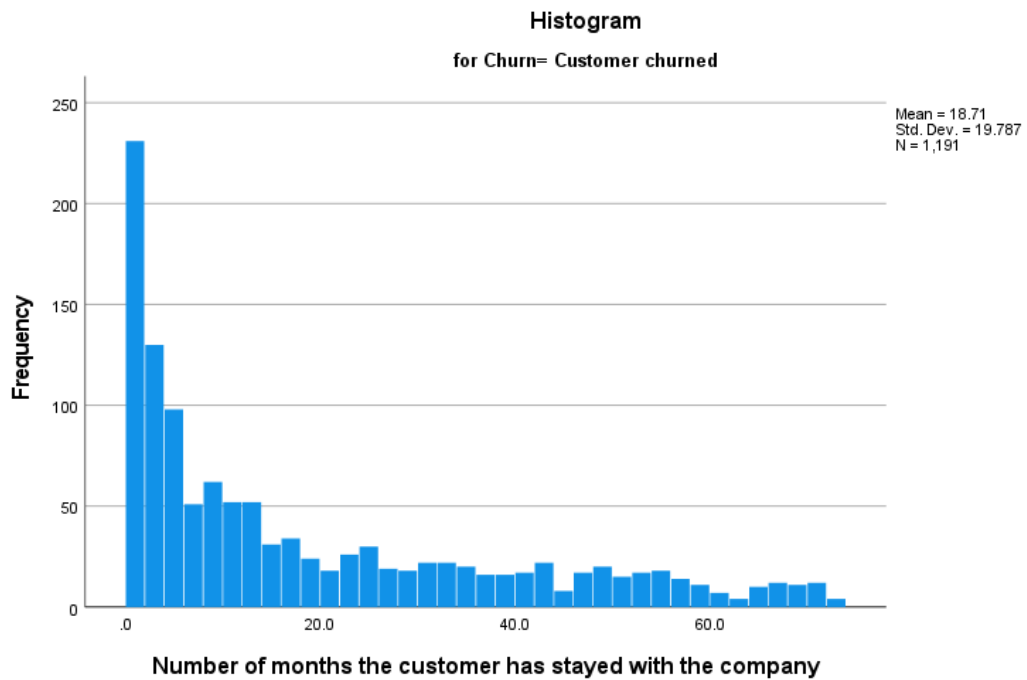
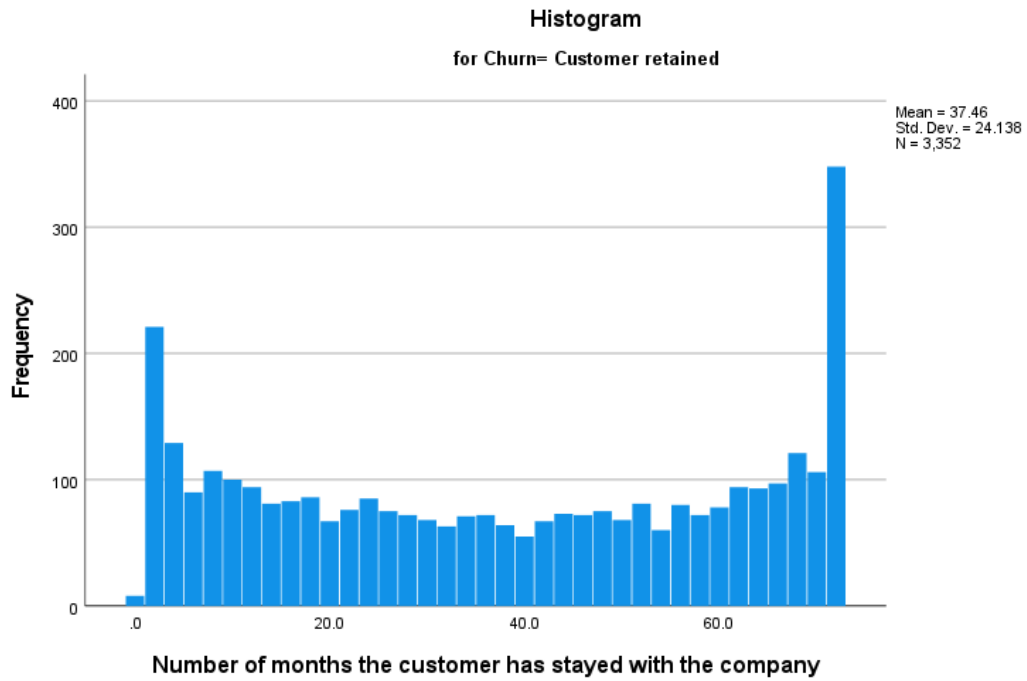
Tests of Normality

Whether the customer has left the company (Yes/No)		Kolmogorov-Smirnov ^a			Shapiro-Wilk		
		Statistic	df	Sig.	Statistic	df	Sig.
Number of months the customer has stayed with the company	Customer retained	.094	3352	.000	.917	3352	.000
	Customer churned	.185	1191	.000	.831	1191	.000

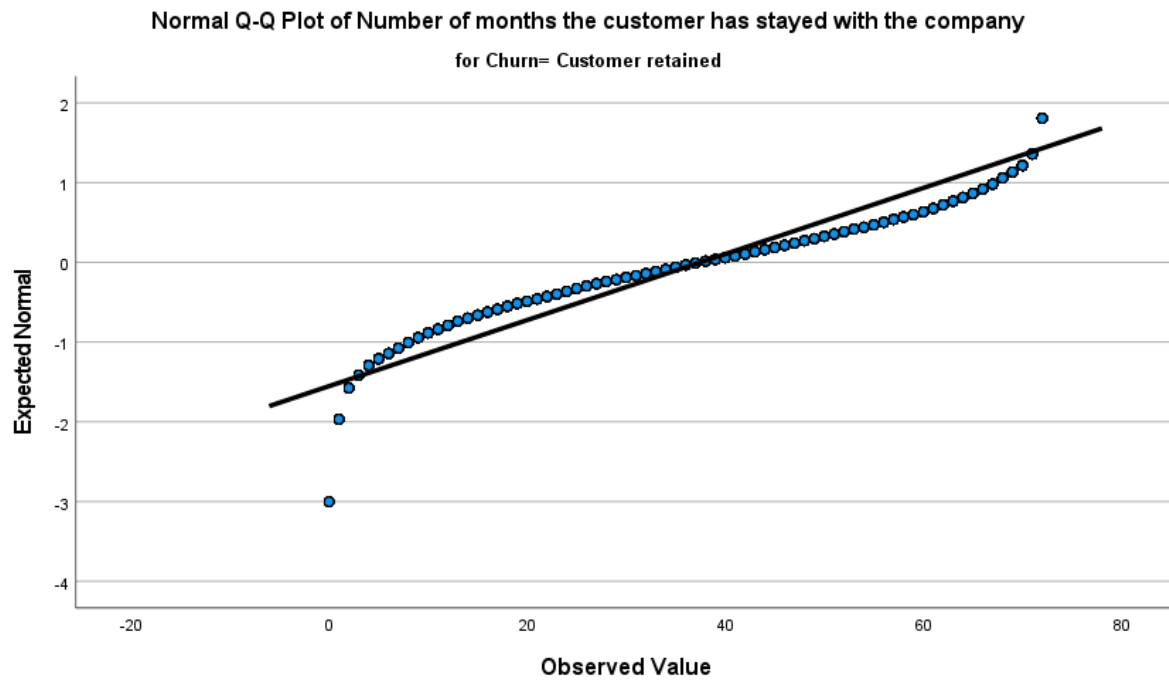
a. Lilliefors Significance Correction

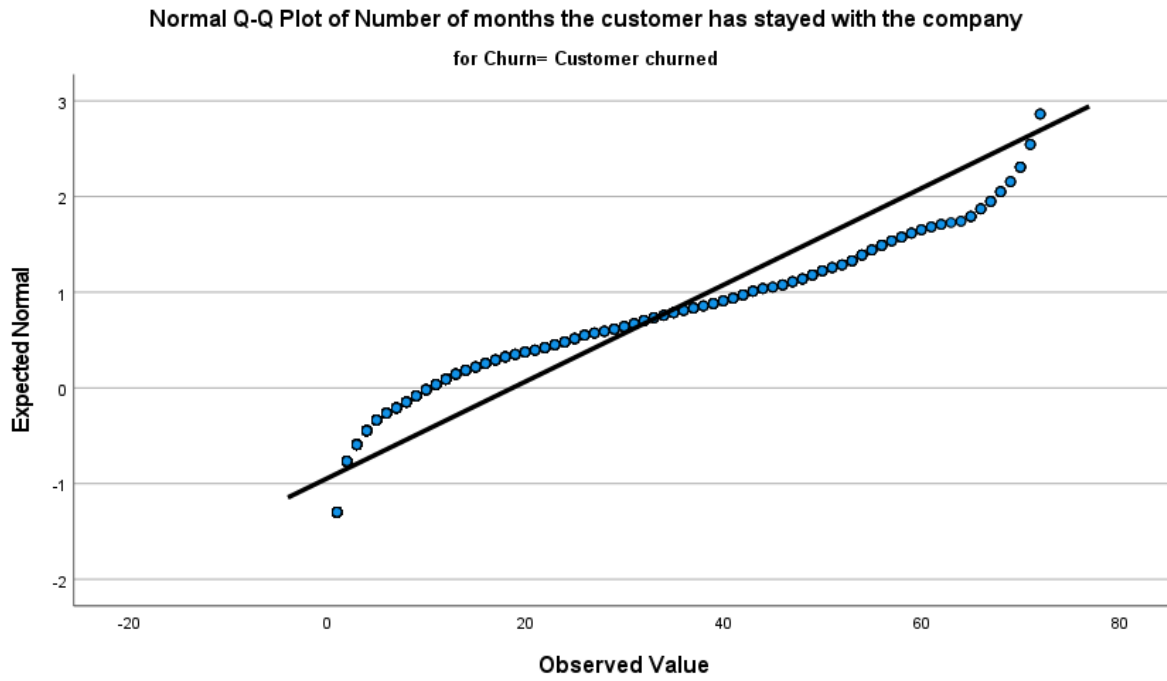
Number of months the customer has stayed with the company

Histograms

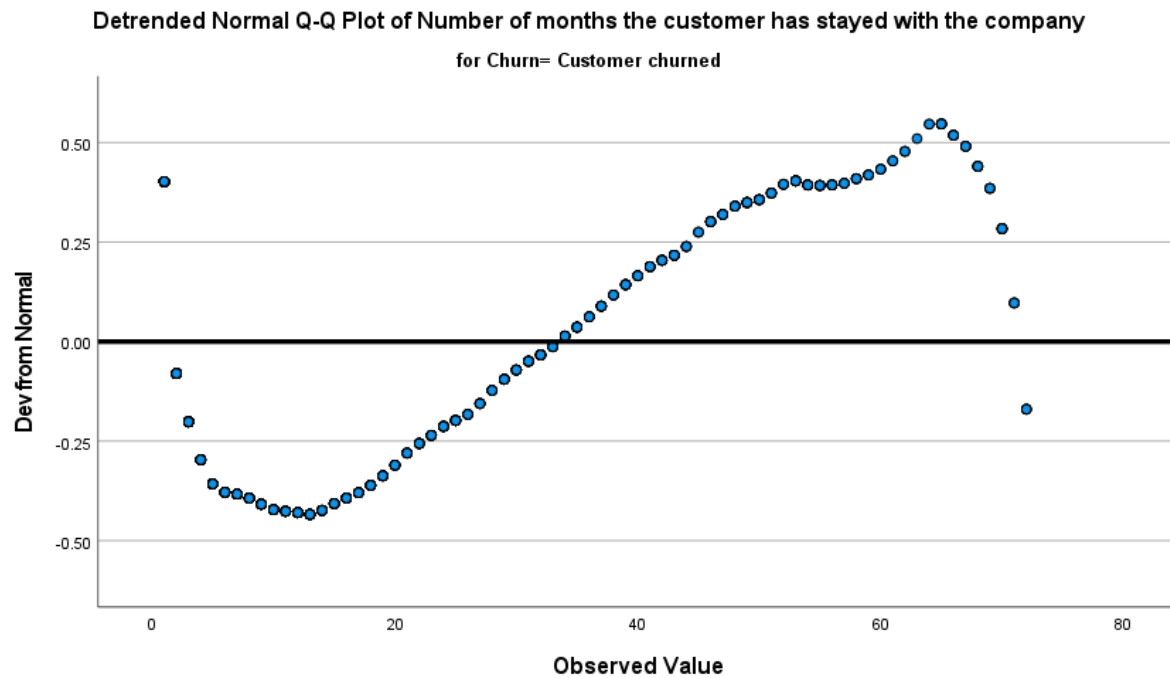
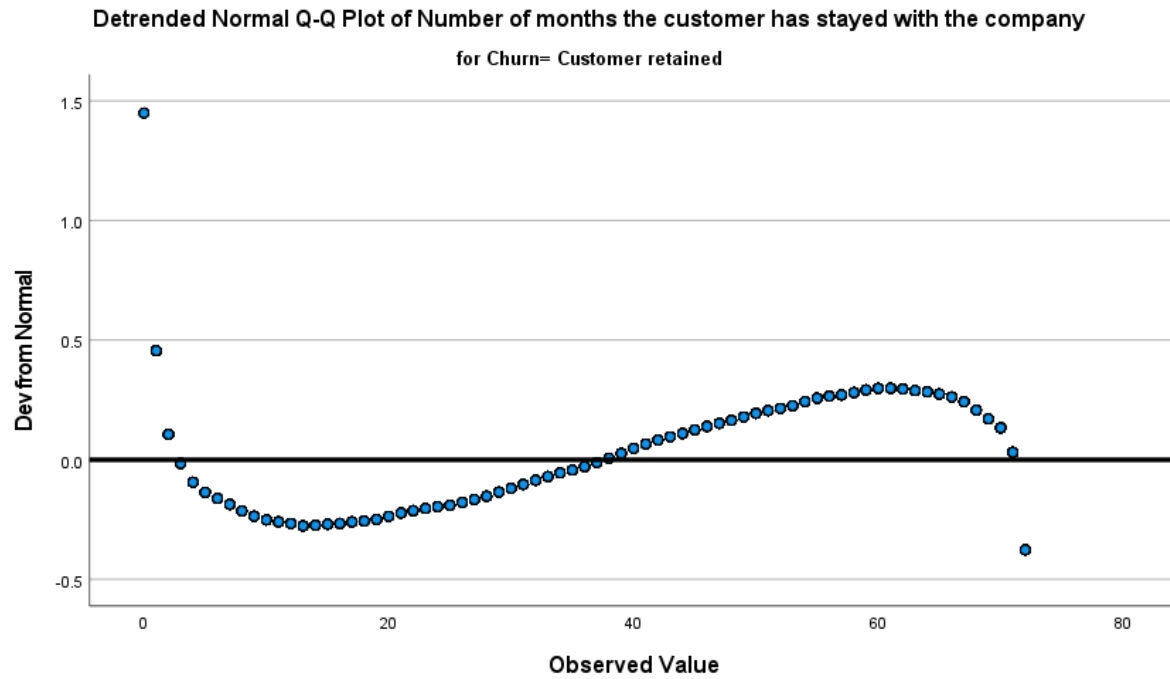


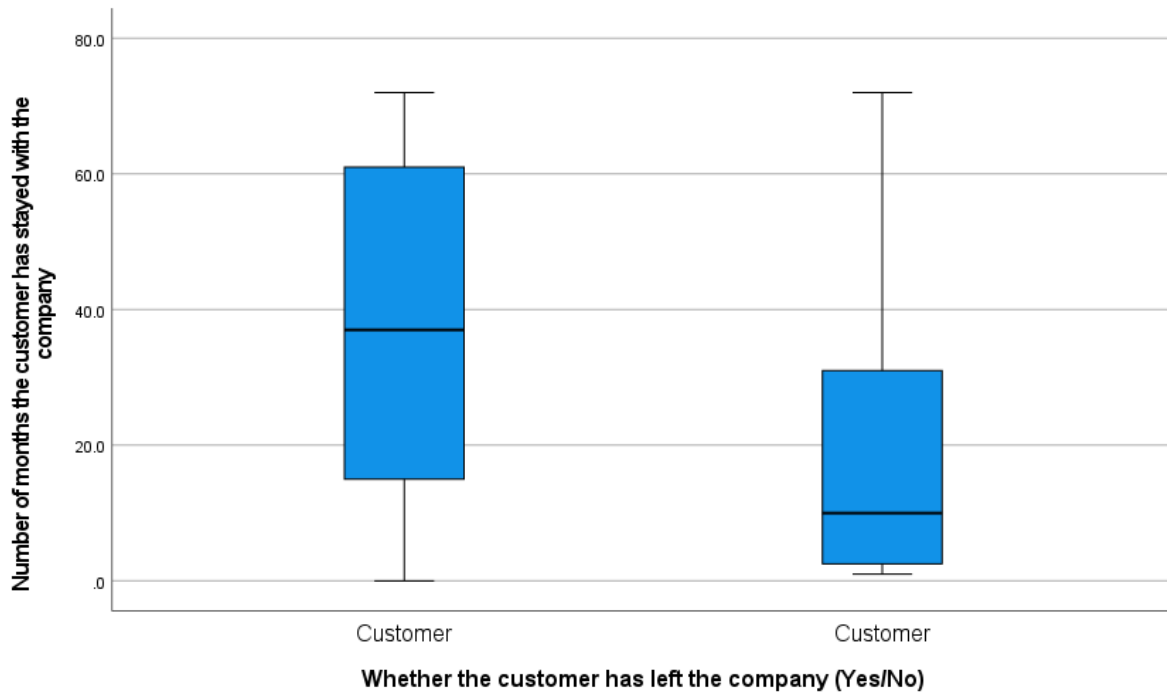
Normal Q-Q Plots





Detrended Normal Q-Q Plots





* Data deviates from normality, moving to non-parametric tests

```
RECODE Churn ('No'=0) ('Yes'=1) INTO Churn_recoded.
```

```
VALUE LABELS Churn_recoded
```

```
0 'No'
```

```
1 'Yes'.
```

```
EXECUTE.
```

```
NPAR TESTS
```

```
/M-W= tenure BY Churn_recoded(0 1)
```

```
/STATISTICS=DESCRIPTIVES QUARTILES
```

```
/MISSING ANALYSIS.
```

NPar Tests

Descriptive Statistics								
	N	Mean	Std. Deviation	Minimum	Maximum	Percentiles		
						25th	50th (Median)	75th
Number of months the customer has stayed with the company	4543	32.547	24.5055	.0	72.0	9.000	29.000	56.000
Churn_recoded	7043	.2654	.44156	.00	1.00	.0000	.0000	1.0000

Mann-Whitney Test

Ranks				
	Churn_recoded	N	Mean Rank	Sum of Ranks
Number of months the customer has	No	3352	2546.27	8535101.00
	Yes	1191	1500.08	1786595.00

stayed with the company	Total	4543		
-------------------------	-------	------	--	--

Test Statistics^a

Number of months the customer has stayed with the company

Mann-Whitney U	1076759.000
Wilcoxon W	1786595.000
Z	-23.656
Asymp. Sig. (2-tailed)	.000

a. Grouping Variable:
Churn_recoded

* /RQ-2: Are contract type and customer churn associated?/

DATASET ACTIVATE DataSet1.

CROSSTABS

/TABLES= Churn BY Contract

/FORMAT=AVALUE TABLES

/STATISTICS=CHISQ PHI

/CELLS=COUNT COLUMN

/COUNT ROUND CELL

/BARChart.

Crosstabs

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Whether the customer has left the company (Yes/No) * Type of contract: Month-to-month, One year, Two year	7043	100.0%	0	0.0%	7043	100.0%

Whether the customer has left the company (Yes/No) * Type of contract: Month-to-month, One year, Two year
Crosstabulation

Type of contract: Month-to-month, One year, Two year							
Month-to-month		One year		Two year		Total	
N	%	N	%	N	%	N	%

Whether the customer has left the company (Yes/No)	Customer retained	2220	57.3%	1307	88.7%	1647	97.2%	5174	73.5%
	Customer churned	1655	42.7%	166	11.3%	48	2.8%	1869	26.5%
Total		3875	100.0%	1473	100.0%	1695	100.0%	7043	100.0%

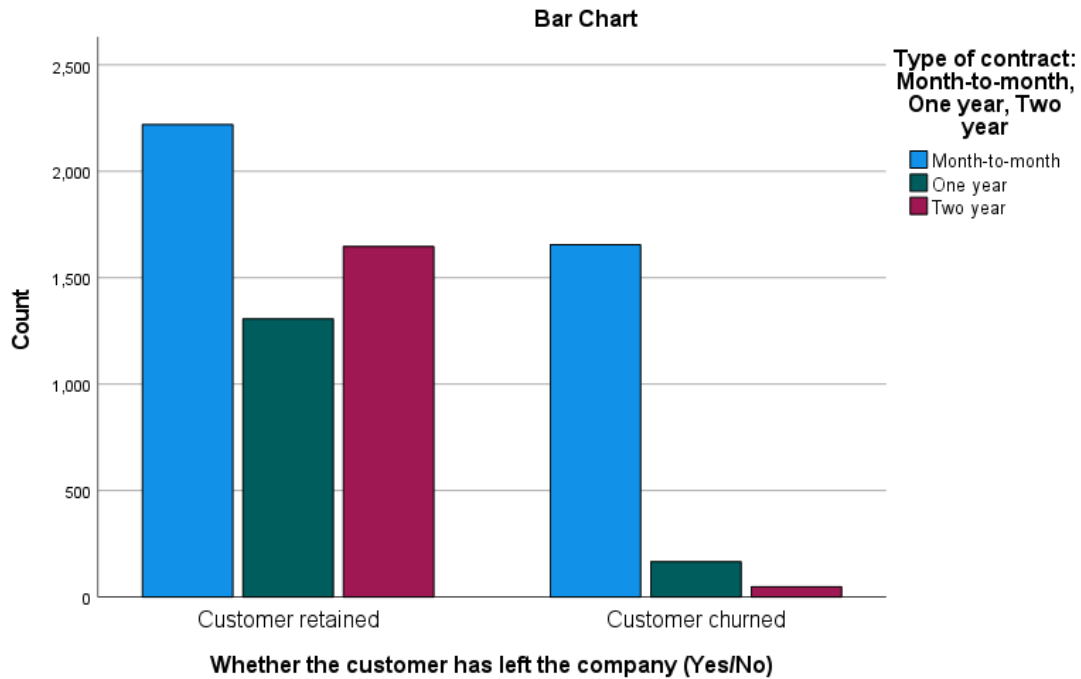
Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	1184.597 ^a	2	.000
Likelihood Ratio	1386.810	2	.000
N of Valid Cases	7043		

a. 0 cells (0.0%) have expected count less than 5.
The minimum expected count is 390.89.

Symmetric Measures

		Value	Approximate Significance
Nominal by Nominal	Phi	.410	.000
	Cramer's V	.410	.000
N of Valid Cases		7043	



OUTPUT MODIFY

/SELECT TABLES

/IF COMMANDS=["Crosstabs(LAST)"] SUBTYPES=["Crosstabulation"]

/TABLE PIVOT=[R1,C1].

OUTPUT MODIFY

/SELECT TABLES

/IF COMMANDS=["Crosstabs(LAST)"] SUBTYPES=["Crosstabulation"]

/TABLECELLS SELECT=[PERCENT] APPLYTO=COLUMNHEADER REPLACE="%"

/TABLECELLS SELECT=[COUNT] APPLYTO=COLUMNHEADER REPLACE="N".

* /RQ-3: Which customer characteristics are associated with the odds of customer churn?/

LOGISTIC REGRESSION VARIABLES Churn_recoded

/METHOD=ENTER tenure MonthlyCharges Contract TechSupport InternetService
PaymentMethod

/CONTRAST (InternetService)=Indicator

/CONTRAST (TechSupport)=Indicator(1)

/CONTRAST (Contract)=Indicator(1)

/CONTRAST (PaymentMethod)=Indicator(1)

/PRINT=GOODFIT CI(95)

/CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).

Logistic Regression

Warnings

Text: Contract Command: LOGISTIC
REGRESSION

This procedure cannot use string variables longer
than 8 bytes. The values will be truncated.

Text: TechSupport Command: LOGISTIC
REGRESSION

This procedure cannot use string variables longer
than 8 bytes. The values will be truncated.

Text: InternetService Command: LOGISTIC
REGRESSION

This procedure cannot use string variables longer
than 8 bytes. The values will be truncated.

Text: PaymentMethod Command: LOGISTIC
REGRESSION

This procedure cannot use string variables longer
than 8 bytes. The values will be truncated.

Due to redundancies, degrees of freedom have
been reduced for one or more variables.

Case Processing Summary

Unweighted Cases ^a		N	Percent
Selected Cases	Included in Analysis	4543	64.5
	Missing Cases	2500	35.5
	Total	7043	100.0
Unselected Cases		0	.0
Total		7043	100.0

a. If weight is in effect, see classification table for
the total number of cases.

Dependent Variable Encoding

Original Value	Internal Value
-------------------	-------------------

No	0
Yes	1

Categorical Variables Codings

		Frequency	Parameter coding		
			(1)	(2)	(3)
Payment method (e.g., Electronic check, Mailed check, etc.)	Bank tra	1012	.000	.000	.000
	Credit c	976	1.000	.000	.000
	Electron	1529	.000	1.000	.000
	Mailed c	1026	.000	.000	1.000
Whether the customer has tech support	No	2225	.000	.000	
	No inter	1008	1.000	.000	
	Yes	1310	.000	1.000	
Customer's internet service provider (DSL, Fiber optic, No)	DSL	1559	1.000	.000	
	Fiber op	1976	.000	1.000	
	No	1008	.000	.000	
Type of contract: Month-to-month, One year, Two year	Month-to	2462	.000	.000	
	One year	981	1.000	.000	
	Two year	1100	.000	1.000	

Block 0: Beginning Block

Classification Table^{a,b}

Observed			Predicted		Percentage Correct
			Churn_recoded No	Yes	
Step 0	Churn_recoded	No	3352	0	100.0
		Yes	1191	0	.0
	Overall Percentage				73.8

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-1.035	.034	940.926	1	.000	.355

Variables not in the Equation^a

			Score	df	Sig.
Step 0	Variables	Number of months the customer has stayed with the company	515.074	1	.000
		Monthly charges	164.711	1	.000

Type of contract: Month-to-month, One year, Two year	743.627	2	.000
Type of contract: Month-to-month, One year, Two year(1)	130.199	1	.000
Type of contract: Month-to-month, One year, Two year(2)	410.795	1	.000
Whether the customer has tech support	495.909	2	.000
Whether the customer has tech support(1)	228.663	1	.000
Whether the customer has tech support(2)	104.739	1	.000
Customer's internet service provider (DSL, Fiber optic, No)	451.596	2	.000
Customer's internet service provider (DSL, Fiber optic, No)(1)	63.001	1	.000
Customer's internet service provider (DSL, Fiber optic, No)(2)	411.094	1	.000
Payment method (e.g., Electronic check, Mailed check, etc.)	386.008	3	.000

	Payment method (e.g., Electronic check, Mailed check, etc.)(1)	77.049	1	.000
	Payment method (e.g., Electronic check, Mailed check, etc.)(2)	383.049	1	.000
	Payment method (e.g., Electronic check, Mailed check, etc.)(3)	39.576	1	.000

a. Residual Chi-Squares are not computed because of redundancies.

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	1312.971	10	.000
	Block	1312.971	10	.000
	Model	1312.971	10	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	3914.257 ^a	.251	.367

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	5.454	8	.708

Contingency Table for Hosmer and Lemeshow Test

		Churn_recoded = No		Churn_recoded = Yes		Total
		Observed	Expected	Observed	Expected	
Step 1	1	451	448.893	3	5.107	454
	2	442	441.890	12	12.110	454
	3	429	430.532	25	23.468	454
	4	408	409.332	46	44.668	454
	5	367	379.402	86	73.598	453
	6	353	347.932	101	106.068	454
	7	311	301.756	143	152.244	454

8	251	253.778	203	200.222	454
9	204	197.205	250	256.795	454
10	136	141.282	322	316.718	458

Classification Table^a

			Predicted		
			Churn_recoded		Percentage Correct
Observed			No	Yes	
Step 1	Churn_recoded	No	3018	334	90.0
		Yes	625	566	47.5
	Overall Percentage				78.9

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
Step 1 ^a	Number of months the customer has stayed with the company	-.028	.003	116.915	1	.000	.973	.968	.977
	Monthly charges	.006	.004	2.688	1	.101	1.006	.999	1.014
	Type of contract: Month-to-month, One year, Two year			78.102	2	.000			
	Type of contract: Month-to-month, One year, Two year(1)	-.777	.126	38.312	1	.000	.460	.359	.588

Type of contract: Month-to-month, One year, Two year(2)	-1.674	.214	61.470	1	.000	.188	.123	.285
Whether the customer has tech support			44.171	2	.000			
Whether the customer has tech support(1)	-1.553	.290	28.664	1	.000	.212	.120	.374
Whether the customer has tech support(2)	-.307	.108	8.059	1	.005	.736	.595	.909
Customer's internet service provider (DSL, Fiber optic, No)			21.540	1	.000			
Customer's internet service provider (DSL, Fiber optic, No)(1)	-.769	.166	21.540	1	.000	.464	.335	.641
Payment method (e.g., Electronic check, Mailed check, etc.)			28.512	3	.000			
Payment method (e.g., Electronic check, Mailed check, etc.)(1)	-.051	.138	.138	1	.710	.950	.726	1.244
Payment method (e.g., Electronic check, Mailed check, etc.)(2)	.394	.113	12.234	1	.000	1.483	1.189	1.850
Payment method (e.g., Electronic check, Mailed check, etc.)(3)	-.096	.137	.487	1	.485	.909	.695	1.189
Constant	.064	.340	.035	1	.852	1.066		

a. Variable(s) entered on step 1: Number of months the customer has stayed with the company, Monthly charges, Type of contract: Month-to-month, One year, Two year, Whether the customer has tech support, Customer's internet service provider (DSL, Fiber optic, No), Payment method (e.g., Electronic check, Mailed check, etc.).

* Due to redundancies (in predictor variables), degrees of freedom have been reduced for one or more variables (from 11 df to 10 df).

* Now we'll test two reduced models to see which one is more usable.

* Model-1 (Uses InternetService):

LOGISTIC REGRESSION VARIABLES Churn_recoded

/METHOD=ENTER tenure MonthlyCharges PaymentMethod Contract InternetService

/CONTRAST (Contract)=Indicator(1)

/CONTRAST (InternetService)=Indicator

/CONTRAST (PaymentMethod)=Indicator(1)

/PRINT=GOODFIT CI(95)

/CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).

Logistic Regression

Warnings

Text: PaymentMethod Command: LOGISTIC
REGRESSION

This procedure cannot use string variables longer
than 8 bytes. The values will be truncated.

Text: Contract Command: LOGISTIC
REGRESSION

This procedure cannot use string variables longer than 8 bytes. The values will be truncated.

Text: InternetService Command: LOGISTIC
REGRESSION

This procedure cannot use string variables longer than 8 bytes. The values will be truncated.

Case Processing Summary

Unweighted Cases ^a		N	Percent
Selected Cases	Included in Analysis	4543	64.5
	Missing Cases	2500	35.5
	Total	7043	100.0
Unselected Cases		0	.0
Total		7043	100.0

a. If weight is in effect, see classification table for the total number of cases.

Dependent Variable Encoding

Original Value	Internal Value
No	0
Yes	1

Categorical Variables Codings

			Parameter coding		
Frequency			(1)	(2)	(3)
Payment method (e.g., Electronic check, Mailed check, etc.)	Bank tra	1012	.000	.000	.000
	Credit c	976	1.000	.000	.000
	Electron	1529	.000	1.000	.000
	Mailed c	1026	.000	.000	1.000
Customer's internet service provider (DSL, Fiber optic, No)	DSL	1559	1.000	.000	
	Fiber op	1976	.000	1.000	
	No	1008	.000	.000	
Type of contract: Month-to-month, One year, Two year	Month-to	2462	.000	.000	
	One year	981	1.000	.000	
	Two year	1100	.000	1.000	

Block 0: Beginning Block

Classification Table^{a,b}

			Predicted		
			Churn_recoded		Percentage Correct
Observed			No	Yes	
Step 0	Churn_recoded	No	3352	0	100.0
		Yes	1191	0	.0
	Overall Percentage				73.8

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-1.035	.034	940.926	1	.000	.355

Variables not in the Equation

			Score	df	Sig.
Step 0	Variables	Number of months the customer has stayed with the company	515.074	1	.000
		Monthly charges	164.711	1	.000
		Payment method (e.g., Electronic check, Mailed check, etc.)	386.008	3	.000

	Payment method (e.g., Electronic check, Mailed check, etc.)(1)	77.049	1	.000
	Payment method (e.g., Electronic check, Mailed check, etc.)(2)	383.049	1	.000
	Payment method (e.g., Electronic check, Mailed check, etc.)(3)	39.576	1	.000
	Type of contract: Month-to-month, One year, Two year	743.627	2	.000
	Type of contract: Month-to-month, One year, Two year(1)	130.199	1	.000
	Type of contract: Month-to-month, One year, Two year(2)	410.795	1	.000
	Customer's internet service provider (DSL, Fiber optic, No)	451.596	2	.000
	Customer's internet service provider (DSL, Fiber optic, No)(1)	63.001	1	.000
	Customer's internet service provider (DSL, Fiber optic, No)(2)	411.094	1	.000

Overall Statistics	1121.635	9	.000
--------------------	----------	---	------

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	1304.851	9	.000
	Block	1304.851	9	.000
	Model	1304.851	9	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	3922.377 ^a	.250	.365

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	17.656	8	.024

Contingency Table for Hosmer and Lemeshow Test

		Churn_recoded = No		Churn_recoded = Yes		Total
		Observed	Expected	Observed	Expected	
Step 1	1	450	448.981	4	5.019	454
	2	446	441.919	8	12.081	454
	3	424	430.365	30	23.635	454
	4	411	408.705	43	45.295	454
	5	358	381.523	98	74.477	456
	6	366	346.556	88	107.444	454
	7	307	302.020	147	151.980	454
	8	250	253.915	204	200.085	454
	9	201	196.698	253	257.302	454
	10	139	141.319	316	313.681	455

Classification Table^a

Observed	Predicted		Percentage Correct
	Churn_recoded		
	No	Yes	

a. The cut value is .500

Type of contract: Month-to-month, One year, Two year(2)	-1.752	.212	68.363	1	.000	.173	.115	.263
Customer's internet service provider (DSL, Fiber optic, No)			39.482	2	.000			
Customer's internet service provider (DSL, Fiber optic, No)(1)	.790	.186	18.113	1	.000	2.204	1.532	3.171
Customer's internet service provider (DSL, Fiber optic, No)(2)	1.710	.284	36.339	1	.000	5.528	3.170	9.637
Constant	-1.399	.175	63.620	1	.000	.247		

a. Variable(s) entered on step 1: Number of months the customer has stayed with the company, Monthly charges, Payment method (e.g., Electronic check, Mailed check, etc.), Type of contract: Month-to-month, One year, Two year, Customer's internet service provider (DSL, Fiber optic, No).

* H-L test is significant, thus this model is rejected.

* Model-2 (Uses TechSupport):

LOGISTIC REGRESSION VARIABLES Churn_recoded

/METHOD=ENTER tenure MonthlyCharges PaymentMethod Contract TechSupport

/CONTRAST (Contract)=Indicator(1)

/CONTRAST (TechSupport)=Indicator(1)

/CONTRAST (PaymentMethod)=Indicator(1)

/PRINT=GOODFIT CI(95)

/CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).

Logistic Regression

Warnings

Text: PaymentMethod Command: LOGISTIC REGRESSION

This procedure cannot use string variables longer than 8 bytes. The values will be truncated.

Text: Contract Command: LOGISTIC REGRESSION

This procedure cannot use string variables longer than 8 bytes. The values will be truncated.

Text: TechSupport Command: LOGISTIC REGRESSION

This procedure cannot use string variables longer than 8 bytes. The values will be truncated.

Case Processing Summary

Unweighted Cases ^a		N	Percent
Selected Cases	Included in Analysis	4543	64.5
	Missing Cases	2500	35.5

Total	7043	100.0
Unselected Cases	0	.0
Total	7043	100.0

a. If weight is in effect, see classification table for the total number of cases.

Dependent Variable Encoding

Original Value	Internal Value
No	0
Yes	1

Categorical Variables Codings

		Frequency	Parameter coding		
			(1)	(2)	(3)
Payment method (e.g., Electronic check, Mailed check, etc.)	Bank tra	1012	.000	.000	.000
	Credit c	976	1.000	.000	.000
	Electron	1529	.000	1.000	.000
	Mailed c	1026	.000	.000	1.000
Whether the customer has tech support	No	2225	.000	.000	
	No inter	1008	1.000	.000	
	Yes	1310	.000	1.000	
	Month-to	2462	.000	.000	

Type of contract:	One	981	1.000	.000	
Month-to-month, One	year				
year, Two year	Two	1100	.000	1.000	
	year				

Block 0: Beginning Block

Classification Table^{a,b}

Observed			Predicted		Percentage Correct
			Churn_recoded		
			No	Yes	
Step 0	Churn_recoded	No	3352	0	100.0
		Yes	1191	0	.0
	Overall Percentage				73.8

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 0	Constant	-1.035	.034	940.926	1	.000	.355

Variables not in the Equation

			Score	df	Sig.
Step 0	Variable s	Number of months the customer has stayed with the company	515.074	1	.000
		Monthly charges	164.711	1	.000
		Payment method (e.g., Electronic check, Mailed check, etc.)	386.008	3	.000
		Payment method (e.g., Electronic check, Mailed check, etc.)(1)	77.049	1	.000
		Payment method (e.g., Electronic check, Mailed check, etc.)(2)	383.049	1	.000
		Payment method (e.g., Electronic check, Mailed check, etc.)(3)	39.576	1	.000
		Type of contract: Month-to-month, One year, Two year	743.627	2	.000
		Type of contract: Month-to-month, One year, Two year(1)	130.199	1	.000
		Type of contract: Month-to-month, One year, Two year(2)	410.795	1	.000

	Whether the customer has tech support	495.909	2	.000
	Whether the customer has tech support(1)	228.663	1	.000
	Whether the customer has tech support(2)	104.739	1	.000
	Overall Statistics	1098.374	9	.000

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	1291.125	9	.000
	Block	1291.125	9	.000
	Model	1291.125	9	.000

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	3936.103 ^a	.247	.362

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	11.632	8	.168

Contingency Table for Hosmer and Lemeshow Test

		Churn_recoded = No		Churn_recoded = Yes		Total
		Observed	Expected	Observed	Expected	
Step 1	1	449	449.191	5	4.809	454
	2	445	441.749	9	12.251	454
	3	429	431.198	25	22.802	454
	4	404	408.933	50	45.067	454
	5	370	378.659	84	75.341	454
	6	346	346.498	108	107.502	454
	7	303	297.167	151	156.833	454
	8	277	250.277	177	203.723	454

9	195	201.492	259	252.508	454
10	134	146.835	323	310.165	457

Classification Table^a

Observed		Predicted		Percentage Correct
		Churn_recoded No	Yes	
Step 1	Churn_recoded No	3004	348	89.6
	Yes	597	594	49.9
	Overall Percentage			79.2

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)	95% C.I. for EXP(B)	
								Lower	Upper
Step 1 ^a	Number of months the customer has stayed with the company	-.029	.003	133.258	1	.000	.971	.966	.976
	Monthly charges	.022	.002	95.177	1	.000	1.022	1.017	1.026
	Payment method (e.g., Electronic check, Mailed check, etc.)			32.633	3	.000			
	Payment method (e.g., Electronic check, Mailed check, etc.)(1)	-.049	.137	.127	1	.721	.952	.728	1.246

Payment method (e.g., Electronic check, Mailed check, etc.)(2)	.412	.112	13.483	1	.000	1.510	1.212	1.881
Payment method (e.g., Electronic check, Mailed check, etc.)(3)	-.120	.136	.781	1	.377	.887	.679	1.158
Type of contract: Month-to- month, One year, Two year			95.580	2	.000			
Type of contract: Month-to- month, One year, Two year(1)	-.869	.124	49.299	1	.000	.419	.329	.534
Type of contract: Month-to- month, One year, Two year(2)	-1.797	.212	71.909	1	.000	.166	.109	.251
Whether the customer has tech support			26.076	2	.000			
Whether the customer has tech support(1)	-.466	.174	7.142	1	.008	.628	.446	.883
Whether the customer has tech support(2)	-.470	.102	21.078	1	.000	.625	.511	.764
Constant	-1.277	.184	47.933	1	.000	.279		

a. Variable(s) entered on step 1: Number of months the customer has stayed with the company, Monthly charges, Payment method (e.g., Electronic check, Mailed check, etc.), Type of contract: Month-to-month, One year, Two year, Whether the customer has tech support.

* H-L test isn't significant, thus this model is accepted as final model.

* /Task-3: Graphs/

* /G-1/

* Chart Builder.

GGRAPH

```
/GRAPHDATASET NAME="graphdataset" VARIABLES=Churn COUNT()[name="COUNT"]  
MISSING=LISTWISE
```

```
REPORTMISSING=NO
```

```
/GRAPHSPEC SOURCE=INLINE
```

```
/FRAME OUTER=NO INNER=NO
```

```
/GRIDLINES XAXIS=NO YAXIS=YES
```

```
/STYLE GRADIENT=NO.
```

BEGIN GPL

```
SOURCE: s=userSource(id("graphdataset"))
```

```
DATA: Churn=col(source(s), name("Churn"), unit.category())
```

```
DATA: COUNT=col(source(s), name("COUNT"))
```

```
GUIDE: axis(dim(1), label("Whether the customer has left the company (Yes/No)"))
```

```
GUIDE: axis(dim(2), label("Percent"))
```

```
GUIDE: text.title(label("Churn status percentage bar chart"))
```

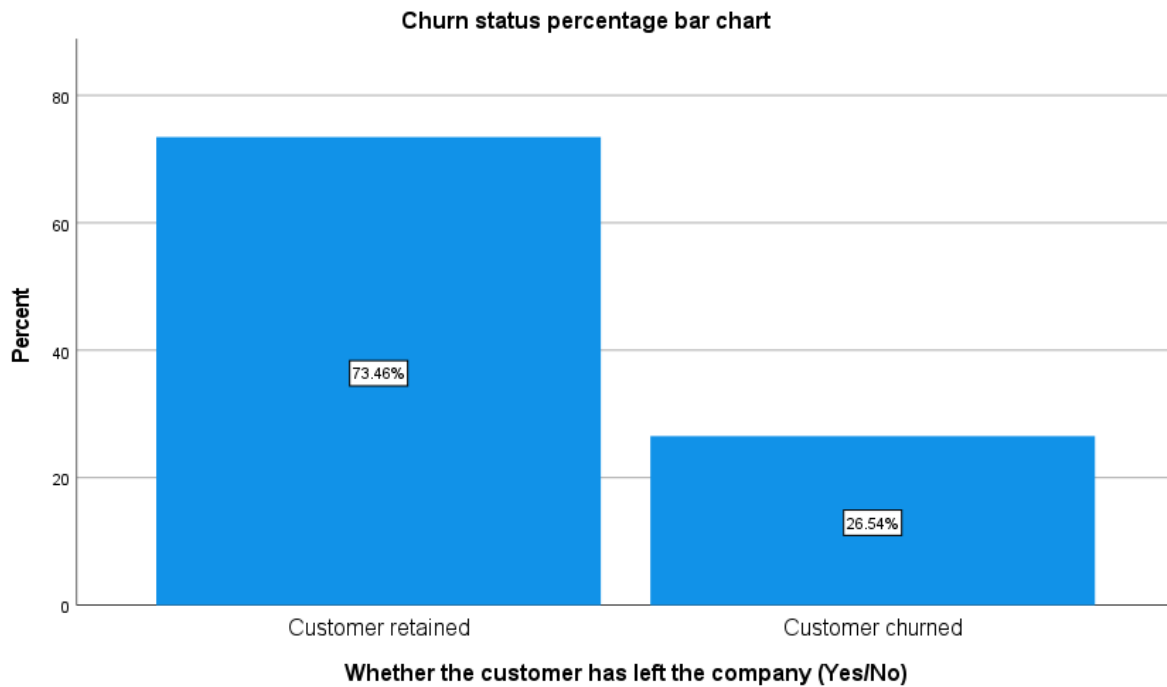
```
SCALE: cat(dim(1), include("No", "Yes"))
```

```
SCALE: linear(dim(2), include(0))
```

```
ELEMENT: interval(position(summary.percent(Churn*COUNT, base.all(acrossPanels()))),  
shape.interior(shape.square))
```

END GPL.

GGraph



* /G-2/

* Chart Builder.

GGRAPH

```
/GRAPHDATASET NAME="graphdataset" VARIABLES=tenure Churn MISSING=LISTWISE  
REPORTMISSING=NO
```

```
/GRAPHSPEC SOURCE=INLINE
```

```
/COLORCYCLE COLOR1(95,195,56), COLOR2(215,0,51), COLOR3(159,24,83),  
COLOR4(250,77,86),
```

COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
 COLOR9(178,134,0),
 COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),
 COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
 COLOR17(208,83,52),
 COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
 COLOR21(28,205,205),
 COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114),
 COLOR25(90,100,94), COLOR26(155,0,0),
 COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
 COLOR30(105,41,196)

/FRAME OUTER=NO INNER=NO
 /GRIDLINES XAXIS=NO YAXIS=YES
 /STYLE GRADIENT=NO.

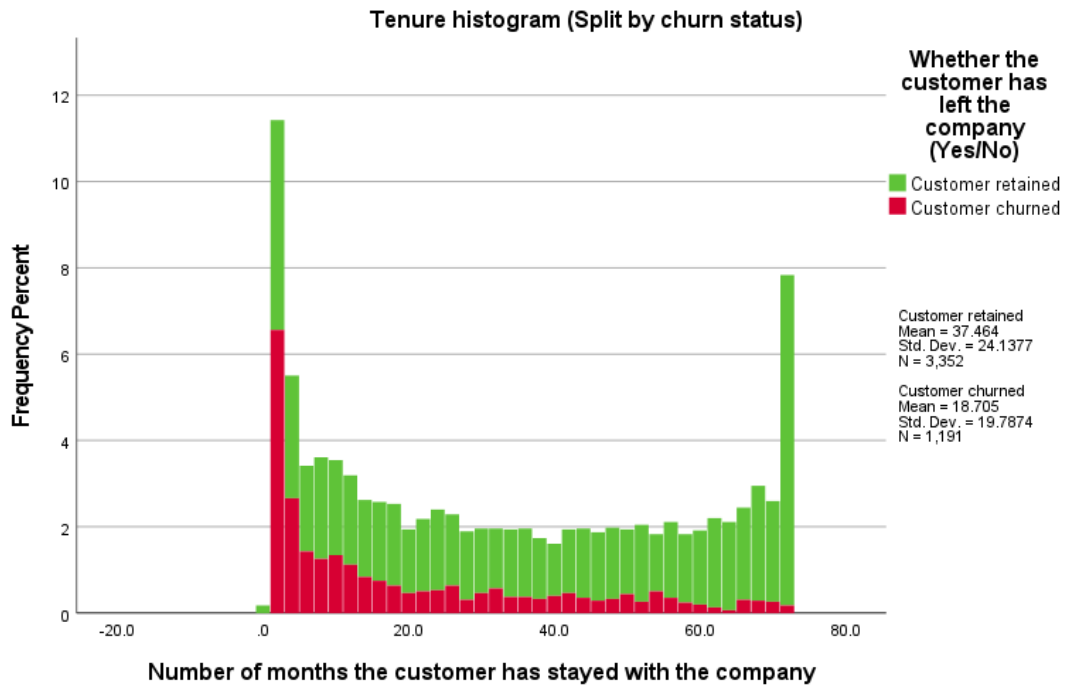
BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))
 DATA: tenure=col(source(s), name("tenure"))
 DATA: Churn=col(source(s), name("Churn"), unit.category())
 GUIDE: axis(dim(1), label("Number of months the customer has stayed with the company"))
 GUIDE: axis(dim(2), label("Frequency Percent"))
 GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left the ",
 "company (Yes/No)"))
 GUIDE: text.title(label("Tenure histogram (Split by churn status)"))
 SCALE: cat(aesthetic(aesthetic.color.interior), include(
 "No", "Yes"))
 ELEMENT: interval.stack(position(summary.percent.count(bin.rect(tenure),

```
base.all(acrossPanels()))), color.interior(Churn), shape.interior(shape.square))
```

END GPL.

GGraph



* /G-3/

* Chart Builder.

GGRAPH

```
/GRAPHDATASET NAME="graphdataset" VARIABLES=Churn tenure MISSING=LISTWISE  
REPORTMISSING=NO
```

```
/GRAPHSPEC SOURCE=INLINE
```

```
/FRAME OUTER=NO INNER=NO
```

```
/GRIDLINES XAXIS=NO YAXIS=YES
```

```
/STYLE GRADIENT=NO.
```

BEGIN GPL

```
SOURCE: s=userSource(id("graphdataset"))
```

```
DATA: Churn=col(source(s), name("Churn"), unit.category())
```

```
DATA: tenure=col(source(s), name("tenure"))
```

```
DATA: id=col(source(s), name("$CASENUM"), unit.category())
```

```
GUIDE: axis(dim(1), label("Whether the customer has left the company (Yes/No)"))
```

```
GUIDE: axis(dim(2), label("Number of months the customer has stayed with the  
company"))
```

```
GUIDE: text.title(label("Tenure by Churn boxplot"))
```

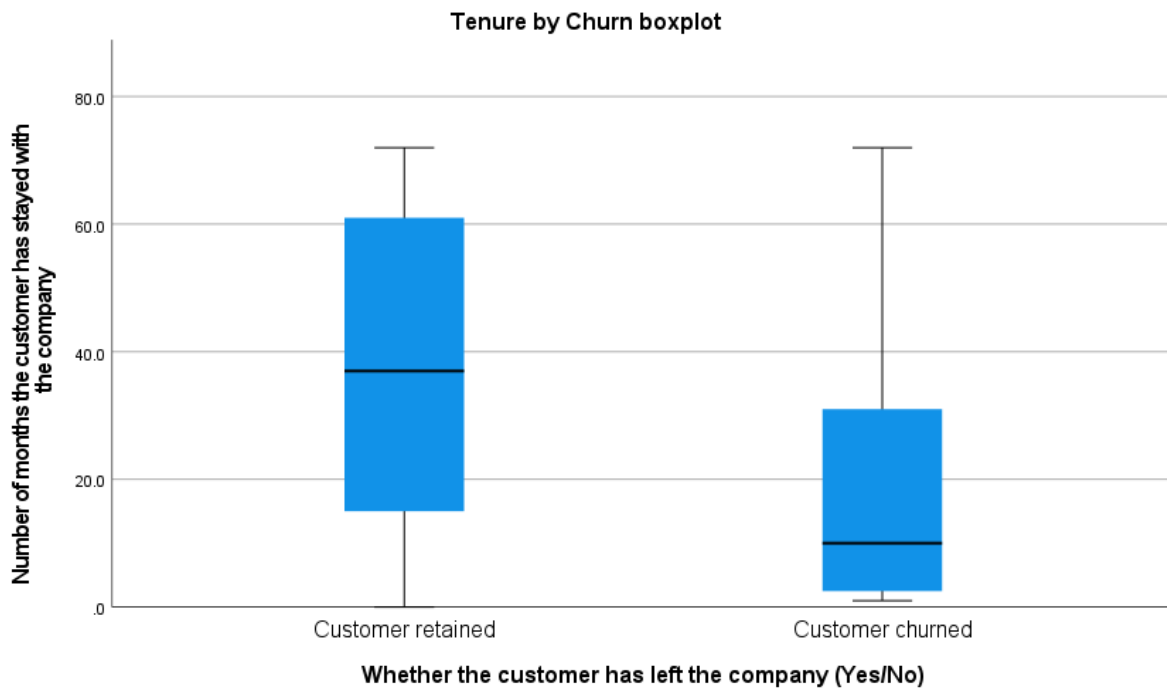
```
SCALE: cat(dim(1), include("No", "Yes"))
```

```
SCALE: linear(dim(2), include(0))
```

```
ELEMENT: schema(position(bin.quantile.letter(Churn*tenure)), label(id))
```

END GPL.

GGraph



* /G-4/

* Chart Builder.

GGRAPH

```
/GRAPHDATASET NAME="graphdataset" VARIABLES=Contract COUNT()[name="COUNT"]
Churn MISSING=LISTWISE
```

```
REPORTMISSING=NO
```

```
/GRAPHSPEC SOURCE=INLINE
```

```
/COLORCYCLE COLOR1(215,0,51), COLOR2(95,195,56), COLOR3(159,24,83),
COLOR4(250,77,86),
```

```
COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
COLOR9(178,134,0),
```

COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),
COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
COLOR17(208,83,52),

COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
COLOR21(28,205,205),

COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114),
COLOR25(90,100,94), COLOR26(155,0,0),

COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
COLOR30(105,41,196)

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: Contract=col(source(s), name("Contract"), unit.category())

DATA: COUNT=col(source(s), name("COUNT"))

DATA: Churn=col(source(s), name("Churn"), unit.category())

COORD: rect(dim(1,2), transpose())

GUIDE: axis(dim(1), label("Type of contract: Month-to-month, One year, Two year"))

GUIDE: axis(dim(2), label("Percent"))

GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left
the ",

"company (Yes/No)"))

GUIDE: text.title(label("Stacked bar chart of contract type by churn status"))

SCALE: cat(dim(1), reverse())

SCALE: linear(dim(2), include(0))

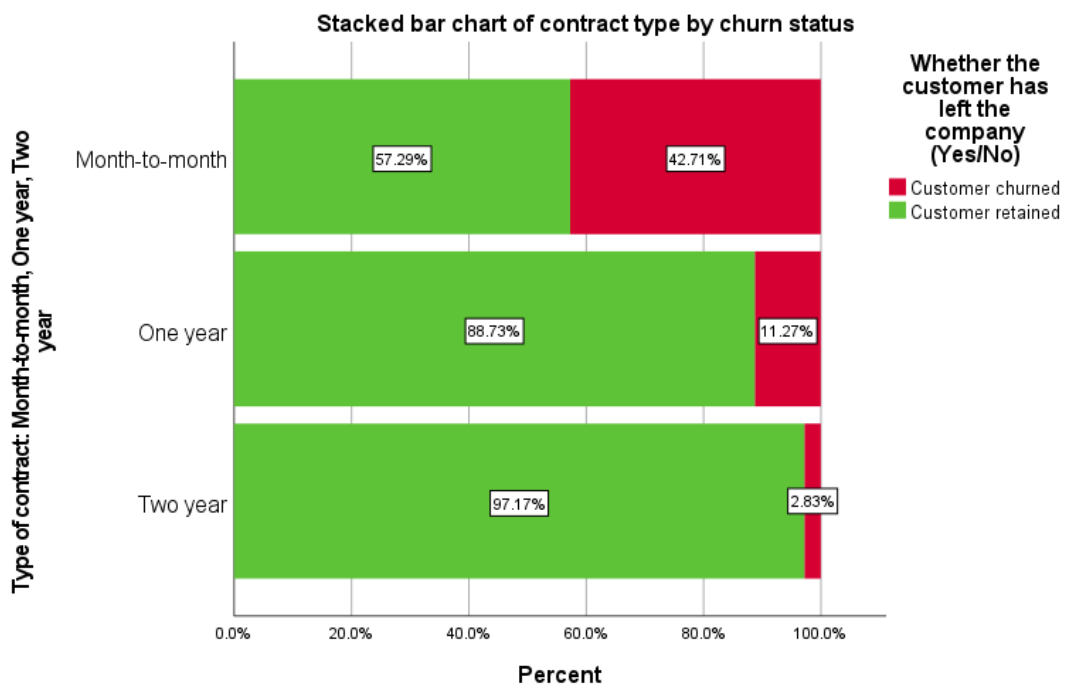
SCALE: cat(aesthetic(aesthetic.color.interior), reverse(), include(

"No", "Yes"))

```
ELEMENT: interval.stack(position(Contract*COUNT), color.interior(Churn),  
  shape.interior(shape.square))
```

END GPL.

GGraph



* /G-5/

* Chart Builder.

GGRAPH

/GRAPHDATASET NAME="graphdataset" VARIABLES=PaymentMethod
COUNT()[name="COUNT"] Churn

MISSING=LISTWISE REPORTMISSING=NO

/GRAPHSPEC SOURCE=INLINE

/COLORCYCLE COLOR1(215,0,51), COLOR2(95,195,56), COLOR3(159,24,83),
COLOR4(250,77,86),

COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
COLOR9(178,134,0),

COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),

COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
COLOR17(208,83,52),

COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
COLOR21(28,205,205),

COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114),
COLOR25(90,100,94), COLOR26(155,0,0),

COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
COLOR30(105,41,196)

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: PaymentMethod=col(source(s), name("PaymentMethod"), unit.category())

DATA: COUNT=col(source(s), name("COUNT"))

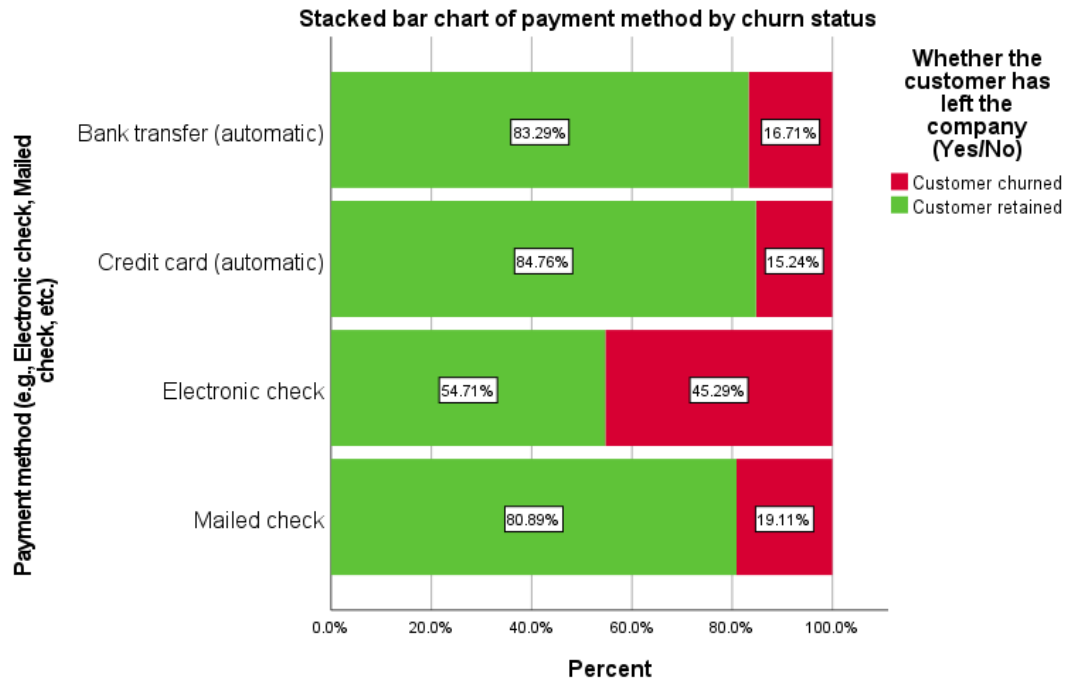
DATA: Churn=col(source(s), name("Churn"), unit.category())

```

COORD: rect(dim(1,2), transpose())
GUIDE: axis(dim(1), label("Payment method (e.g., Electronic check, Mailed check, etc.)"))
GUIDE: axis(dim(2), label("Percent"))
GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left
the ",
  "company (Yes/No)"))
GUIDE: text.title(label("Stacked bar chart of payment method by churn status"))
SCALE: cat(dim(1), reverse())
SCALE: linear(dim(2), include(0))
SCALE: cat(aesthetic(aesthetic.color.interior), reverse(), include(
"No", "Yes"))
ELEMENT: interval.stack(position(PaymentMethod*COUNT), color.interior(Churn),
  shape.interior(shape.square))
END GPL.

```

GGraph



* /G-6/

* Chart Builder.

GGRAPH

```
/GRAPHDATASET NAME="graphdataset" VARIABLES=InternetService
COUNT()[name="COUNT"] Churn
```

```
MISSING=LISTWISE REPORTMISSING=NO
```

```
/GRAPHSPEC SOURCE=INLINE
```

```
/COLORCYCLE COLOR1(215,0,51), COLOR2(95,195,56), COLOR3(159,24,83),
COLOR4(250,77,86),
```

```
COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
COLOR9(178,134,0),
```

```
COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),
```

COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
COLOR17(208,83,52),

COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
COLOR21(28,205,205),

COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114),
COLOR25(90,100,94), COLOR26(155,0,0),

COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
COLOR30(105,41,196)

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: InternetService=col(source(s), name("InternetService"), unit.category())

DATA: COUNT=col(source(s), name("COUNT"))

DATA: Churn=col(source(s), name("Churn"), unit.category())

COORD: rect(dim(1,2), transpose())

GUIDE: axis(dim(1), label("Customer's internet service provider (DSL, Fiber optic, No)"))

GUIDE: axis(dim(2), label("Percent"))

GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left
the ",

"company (Yes/No)"))

GUIDE: text.title(label("Stacked bar chart of customer's internet service by churn status"))

SCALE: linear(dim(2), include(0))

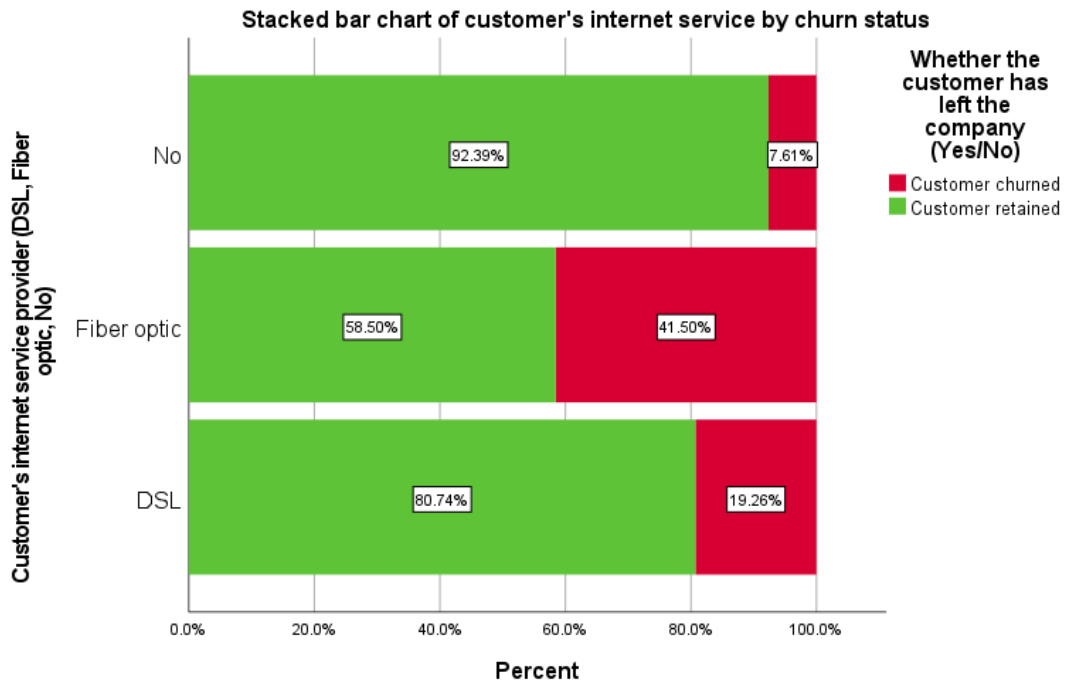
SCALE: cat(aesthetic(aesthetic.color.interior), reverse(), include(
"No", "Yes"))

ELEMENT: interval.stack(position(InternetService*COUNT), color.interior(Churn),

```
shape.interior(shape.square))
```

```
END GPL.
```

GGraph



```
SAVE OUTFILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final  
Project\Final_Project.sav'
```

```
/COMPRESSED.
```

Appendix C

SPSS Syntax

* Encoding: UTF-8.

* /Step-1: Data loading/

PRESERVE.

SET DECIMAL DOT.

GET DATA /TYPE=TXT

/FILE="E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final Project\telco_data.csv"

/ENCODING='UTF8'

/DELIMITERS=","

/QUALIFIER=""

/ARRANGEMENT=DELIMITED

/FIRSTCASE=2

/DATATYPEMIN PERCENTAGE=95.0

/VARIABLES=

customerID AUTO

gender AUTO

SeniorCitizen AUTO

Partner AUTO

Dependents AUTO

tenure AUTO

PhoneService AUTO

MultipleLines AUTO

InternetService AUTO

OnlineSecurity AUTO

OnlineBackup AUTO

DeviceProtection AUTO

TechSupport AUTO

StreamingTV AUTO

StreamingMovies AUTO

Contract AUTO

PaperlessBilling AUTO

PaymentMethod AUTO

MonthlyCharges AUTO

TotalCharges AUTO

Churn AUTO

/MAP.

RESTORE.

CACHE.

EXECUTE.

DATASET NAME DataSet1 WINDOW=FRONT.

SAVE OUTFILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final
Project\Final_Project.sav'

/COMPRESSED.

* /Step-2: Initial data Checking/

DATASET ACTIVATE DataSet1.

FREQUENCIES VARIABLES=customerID gender SeniorCitizen Partner Dependents tenure
PhoneService

MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport
StreamingTV

StreamingMovies Contract PaperlessBilling PaymentMethod MonthlyCharges TotalCharges
Churn

/FORMAT=NOTABLE

/STATISTICS=RANGE MINIMUM MAXIMUM

/ORDER=ANALYSIS.

* /Step-3: Fixing "SPSS failing to find missing value properly" issue/

MISSING VALUES gender Partner InternetService StreamingTV (' ').

* Checking again if things are okay or not

DATASET ACTIVATE DataSet1.

FREQUENCIES VARIABLES=customerID gender SeniorCitizen Partner Dependents tenure
PhoneService

MultipleLines InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport
StreamingTV

StreamingMovies Contract PaperlessBilling PaymentMethod MonthlyCharges TotalCharges
Churn

/FORMAT=NOTABLE

/STATISTICS=RANGE MINIMUM MAXIMUM

/ORDER=ANALYSIS.

* /Step-4: Checking or fixing duplicate observations/

DATASET ACTIVATE DataSet1.

* Identify Duplicate Cases.

SORT CASES BY customerID(A).

MATCH FILES

/FILE=*

/BY customerID

/FIRST=PrimaryFirst

/LAST=PrimaryLast.

DO IF (PrimaryFirst).

COMPUTE MatchSequence=1-PrimaryLast.

ELSE.

COMPUTE MatchSequence=MatchSequence+1.

END IF.

LEAVE MatchSequence.

FORMATS MatchSequence (f7).

COMPUTE InDupGrp=MatchSequence>0.

SORT CASES InDupGrp(D).

MATCH FILES

/FILE=*

/DROP=PrimaryFirst InDupGrp MatchSequence.

VARIABLE LABELS PrimaryLast 'Indicator of each last matching case as Primary'.

VALUE LABELS PrimaryLast 0 'Duplicate Case' 1 'Primary Case'.

VARIABLE LEVEL PrimaryLast (ORDINAL).

FREQUENCIES VARIABLES=PrimaryLast.

EXECUTE.

* No duplicates exist, splendid.

* /Step-4: Data labelling/

VARIABLE LABELS

customerID	"Unique identifier for each customer"
gender	"Customer gender: Male/Female"
SeniorCitizen	"Indicates if the customer is a senior citizen (0 = No, 1 = Yes)"
Partner	"Whether the customer has a partner"
Dependents	"Whether the customer has dependents"
tenure	"Number of months the customer has stayed with the company"
PhoneService	"Whether the customer has phone service"
MultipleLines	"Whether the customer has multiple lines"
InternetService	"Customer's internet service provider (DSL, Fiber optic, No)"
OnlineSecurity	"Whether the customer has online security"
OnlineBackup	"Whether the customer has online backup"
DeviceProtection	"Whether the customer has device protection"
TechSupport	"Whether the customer has tech support"
StreamingTV	"Whether the customer has streaming TV"
StreamingMovies	"Whether the customer has streaming movies"
Contract	"Type of contract: Month-to-month, One year, Two year"
PaperlessBilling	"Whether the customer uses paperless billing"
PaymentMethod	"Payment method (e.g., Electronic check, Mailed check, etc.)"
MonthlyCharges	"Monthly charges"
TotalCharges	"Total charges to date"
Churn	"Whether the customer has left the company (Yes/No)".

EXECUTE.

VALUE LABELS

SeniorCitizen

0 "No"

1 "Yes".

VALUE LABELS

Churn

"No" "Customer retained"

"Yes" "Customer churned".

EXECUTE.

* Things are okay now, we can move onto any required further processing and analysis.

SAVE OUTFILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final
Project\Final_Project.sav'

/COMPRESSED.

* /Task-1: Table-1/

* Custom Tables.

CTABLES

/VLABELS VARIABLES=gender SeniorCitizen Partner Dependents tenure PhoneService
MultipleLines

InternetService OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV
StreamingMovies Contract PaperlessBilling PaymentMethod MonthlyCharges TotalCharges
Churn

DISPLAY=LABEL

/TABLE gender [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] +
SeniorCitizen

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + Partner [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + Dependents

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + tenure [MEAN, STDDEV, MEDIAN,

MINIMUM, MAXIMUM] + PhoneService [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] +

MultipleLines [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + InternetService

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + OnlineSecurity [COUNT 'Frequency'

F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + OnlineBackup [COUNT 'Frequency' F40.0, TABLEPCT.COUNT

'Percentage' PCT40.2] + DeviceProtection [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage'

PCT40.2] + TechSupport [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + StreamingTV

[COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + StreamingMovies [COUNT 'Frequency'

F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] + Contract [COUNT 'Frequency' F40.0, TABLEPCT.COUNT

'Percentage' PCT40.2] + PaperlessBilling [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage'

PCT40.2] + PaymentMethod [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2] +

MonthlyCharges [MEAN, STDDEV, MEDIAN, MINIMUM, MAXIMUM] + TotalCharges [MEAN, STDDEV, MEDIAN,

MINIMUM, MAXIMUM] + Churn [COUNT 'Frequency' F40.0, TABLEPCT.COUNT 'Percentage' PCT40.2]

/CATEGORIES VARIABLES=gender Partner Dependents PhoneService MultipleLines InternetService

OnlineSecurity OnlineBackup DeviceProtection TechSupport StreamingTV StreamingMovies Contract

```
PaperlessBilling PaymentMethod ORDER=A KEY=VALUE EMPTY=EXCLUDE  
/CATEGORIES VARIABLES=SeniorCitizen Churn ORDER=A KEY=VALUE  
EMPTY=INCLUDE  
/CRITERIA CILEVEL=95.
```

* /Task-2: Testing research questions/

* /RQ-1: Is customer tenure associated with customer churn?/

* Testing normality

```
EXAMINE VARIABLES=tenure BY Churn  
/PLOT BOXPLOT HISTOGRAM NPLOT  
/COMPARE GROUPS  
/STATISTICS DESCRIPTIVES  
/CINTERVAL 95  
/MISSING LISTWISE  
/NOTOTAL.
```

* Data deviates from normality, moving to non-parametric tests

```
RECODE Churn ('No'=0) ('Yes'=1) INTO Churn_recoded.  
VALUE LABELS Churn_recoded  
0 'No'  
1 'Yes'.  
EXECUTE.
```

NPART TESTS

/M-W= tenure BY Churn_recoded(0 1)

/STATISTICS=DESCRIPTIVES QUANTILES

/MISSING ANALYSIS.

* /RQ-2: Are contract type and customer churn associated?/

DATASET ACTIVATE DataSet1.

CROSSTABS

/TABLES= Churn BY Contract

/FORMAT=AVALUE TABLES

/STATISTICS=CHISQ PHI

/CELLS=COUNT COLUMN

/COUNT ROUND CELL

/BARCHART.

OUTPUT MODIFY

/SELECT TABLES

/IF COMMANDS=["Crosstabs(LAST)"] SUBTYPES=["Crosstabulation"]

/TABLE PIVOT=[R1,C1].

OUTPUT MODIFY

/SELECT TABLES

/IF COMMANDS=["Crosstabs(LAST)"] SUBTYPES=["Crosstabulation"]

/TABLECELLS SELECT=[PERCENT] APPLYTO=COLUMNHEADER REPLACE="%"

/TABLECELLS SELECT=[COUNT] APPLYTO=COLUMNHEADER REPLACE="N".

* /RQ-3: Which customer characteristics are associated with the odds of customer churn?/

LOGISTIC REGRESSION VARIABLES Churn_recoded

/METHOD=ENTER tenure MonthlyCharges Contract TechSupport InternetService
PaymentMethod

/CONTRAST (InternetService)=Indicator

/CONTRAST (TechSupport)=Indicator(1)

/CONTRAST (Contract)=Indicator(1)

/CONTRAST (PaymentMethod)=Indicator(1)

/PRINT=GOODFIT CI(95)

/CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).

* Due to redundancies (in predictor variables), degrees of freedom have been reduced for one or more variables (from 11 df to 10 df).

* Now we'll test two reduced models to see which one is more usable.

* Model-1 (Uses InternetService):

LOGISTIC REGRESSION VARIABLES Churn_recoded

/METHOD=ENTER tenure MonthlyCharges PaymentMethod Contract InternetService

/CONTRAST (Contract)=Indicator(1)

/CONTRAST (InternetService)=Indicator

/CONTRAST (PaymentMethod)=Indicator(1)

/PRINT=GOODFIT CI(95)

/CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).

* H-L test is significant, thus this model is rejected.

* Model-2 (Uses TechSupport):

LOGISTIC REGRESSION VARIABLES Churn_recoded

/METHOD=ENTER tenure MonthlyCharges PaymentMethod Contract TechSupport

/CONTRAST (Contract)=Indicator(1)

/CONTRAST (TechSupport)=Indicator(1)

/CONTRAST (PaymentMethod)=Indicator(1)

/PRINT=GOODFIT CI(95)

/CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).

* H-L test isn't significant, thus this model is accepted as final model.

* /Task-3: Graphs/

* /G-1/

* Chart Builder.

GGRAPH

/GRAPHDATASET NAME="graphdataset" VARIABLES=Churn COUNT()[name="COUNT"]
MISSING=LISTWISE

REPORTMISSING=NO

/GRAPHSPEC SOURCE=INLINE

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: Churn=col(source(s), name("Churn"), unit.category())

DATA: COUNT=col(source(s), name("COUNT"))

```
GUIDE: axis(dim(1), label("Whether the customer has left the company (Yes/No)))
GUIDE: axis(dim(2), label("Percent"))
GUIDE: text.title(label("Churn status percentage bar chart"))
SCALE: cat(dim(1), include("No", "Yes"))
SCALE: linear(dim(2), include(0))
ELEMENT: interval(position(summary.percent(Churn*COUNT, base.all(acrossPanels()))),
  shape.interior(shape.square))
END GPL.
```

* /G-2/

* Chart Builder.

GGRAPH

```
/GRAPHDATASET NAME="graphdataset" VARIABLES=tenure Churn
MISSING=LISTWISE REPORTMISSING=NO
```

```
/GRAPHSPEC SOURCE=INLINE
```

```
/COLORCYCLE COLOR1(95,195,56), COLOR2(215,0,51), COLOR3(159,24,83),
COLOR4(250,77,86),
```

```
COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
COLOR9(178,134,0),
```

```
COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),
COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
COLOR17(208,83,52),
```

```
COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
COLOR21(28,205,205),
```

```
COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114), COLOR25(90,100,94),
COLOR26(155,0,0),
```

```
COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
COLOR30(105,41,196)
```

```

/FRAME OUTER=NO INNER=NO
/GRIDLINES XAXIS=NO YAXIS=YES
/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))
DATA: tenure=col(source(s), name("tenure"))
DATA: Churn=col(source(s), name("Churn"), unit.category())
GUIDE: axis(dim(1), label("Number of months the customer has stayed with the company"))
GUIDE: axis(dim(2), label("Frequency Percent"))
GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left the ",
  "company (Yes/No)"))
GUIDE: text.title(label("Tenure histogram (Split by churn status)"))
SCALE: cat(aesthetic(aesthetic.color.interior), include(
"No", "Yes"))
ELEMENT: interval.stack(position(summary.percent.count(bin.rect(tenure),
  base.all(acrossPanels()))), color.interior(Churn), shape.interior(shape.square))

END GPL.

```

* /G-3/

* Chart Builder.

GGRAPH

```

/GRAPHDATASET NAME="graphdataset" VARIABLES=Churn tenure
MISSING=LISTWISE REPORTMISSING=NO

/GRAPHSPEC SOURCE=INLINE

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

```

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: Churn=col(source(s), name("Churn"), unit.category())

DATA: tenure=col(source(s), name("tenure"))

DATA: id=col(source(s), name("\$CASENUM"), unit.category())

GUIDE: axis(dim(1), label("Whether the customer has left the company (Yes/No)"))

GUIDE: axis(dim(2), label("Number of months the customer has stayed with the company"))

GUIDE: text.title(label("Tenure by Churn boxplot"))

SCALE: cat(dim(1), include("No", "Yes"))

SCALE: linear(dim(2), include(0))

ELEMENT: schema(position(bin.quantile.letter(Churn*tenure)), label(id))

END GPL.

* /G-4/

* Chart Builder.

GGGRAPH

/GRAPHDATASET NAME="graphdataset" VARIABLES=Contract
COUNT()[name="COUNT"] Churn MISSING=LISTWISE

REPORTMISSING=NO

/GRAPHSPEC SOURCE=INLINE

/COLORCYCLE COLOR1(215,0,51), COLOR2(95,195,56), COLOR3(159,24,83),
COLOR4(250,77,86),

COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
COLOR9(178,134,0),

COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),

COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
COLOR17(208,83,52),

COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
COLOR21(28,205,205),

COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114), COLOR25(90,100,94),
COLOR26(155,0,0),

COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
COLOR30(105,41,196)

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: Contract=col(source(s), name("Contract"), unit.category())

DATA: COUNT=col(source(s), name("COUNT"))

DATA: Churn=col(source(s), name("Churn"), unit.category())

COORD: rect(dim(1,2), transpose())

GUIDE: axis(dim(1), label("Type of contract: Month-to-month, One year, Two year"))

GUIDE: axis(dim(2), label("Percent"))

GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left the ",
"company (Yes/No)"))

GUIDE: text.title(label("Stacked bar chart of contract type by churn status"))

SCALE: cat(dim(1), reverse())

SCALE: linear(dim(2), include(0))

SCALE: cat(aesthetic(aesthetic.color.interior), reverse(), include(
"No", "Yes"))

ELEMENT: interval.stack(position(Contract*COUNT), color.interior(Churn),
shape.interior(shape.square))

END GPL.

* /G-5/

* Chart Builder.

GGRAPH

/GRAPHDATASET NAME="graphdataset" VARIABLES=PaymentMethod
COUNT()[name="COUNT"] Churn

MISSING=LISTWISE REPORTMISSING=NO

/GRAPHSPEC SOURCE=INLINE

/COLORCYCLE COLOR1(215,0,51), COLOR2(95,195,56), COLOR3(159,24,83),
COLOR4(250,77,86),

COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
COLOR9(178,134,0),

COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),

COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
COLOR17(208,83,52),

COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
COLOR21(28,205,205),

COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114), COLOR25(90,100,94),
COLOR26(155,0,0),

COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
COLOR30(105,41,196)

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: PaymentMethod=col(source(s), name("PaymentMethod"), unit.category())

```

DATA: COUNT=col(source(s), name("COUNT"))
DATA: Churn=col(source(s), name("Churn"), unit.category())
COORD: rect(dim(1,2), transpose())
GUIDE: axis(dim(1), label("Payment method (e.g., Electronic check, Mailed check, etc.)"))
GUIDE: axis(dim(2), label("Percent"))
GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left the ",
    "company (Yes/No)"))
GUIDE: text.title(label("Stacked bar chart of payment method by churn status"))
SCALE: cat(dim(1), reverse())
SCALE: linear(dim(2), include(0))
SCALE: cat(aesthetic(aesthetic.color.interior), reverse(), include(
    "No", "Yes"))
ELEMENT: interval.stack(position(PaymentMethod*COUNT), color.interior(Churn),
    shape.interior(shape.square))
END GPL.

```

* /G-6/

* Chart Builder.

GGRAPH

```

/GRAPHDATASET NAME="graphdataset" VARIABLES=InternetService
COUNT()[name="COUNT"] Churn

```

```

MISSING=LISTWISE REPORTMISSING=NO

```

```

/GRAPHSPEC SOURCE=INLINE

```

```

/COLORCYCLE COLOR1(215,0,51), COLOR2(95,195,56), COLOR3(159,24,83),
COLOR4(250,77,86),

```

```

    COLOR5(87,4,8), COLOR6(25,128,56), COLOR7(0,45,156), COLOR8(238,83,139),
COLOR9(178,134,0),

```

```

    COLOR10(0,157,154), COLOR11(1,39,73), COLOR12(138,56,0), COLOR13(165,110,255),
    COLOR14(236,230,208), COLOR15(69,70,71), COLOR16(92,202,136),
    COLOR17(208,83,52),
    COLOR18(204,127,228), COLOR19(225,188,29), COLOR20(237,75,75),
    COLOR21(28,205,205),
    COLOR22(92,113,72), COLOR23(225,139,14), COLOR24(9,38,114), COLOR25(90,100,94),
    COLOR26(155,0,0),
    COLOR27(207,172,227), COLOR28(150,145,145), COLOR29(63,235,124),
    COLOR30(105,41,196)

/FRAME OUTER=NO INNER=NO

/GRIDLINES XAXIS=NO YAXIS=YES

/STYLE GRADIENT=NO.

BEGIN GPL

SOURCE: s=userSource(id("graphdataset"))

DATA: InternetService=col(source(s), name("InternetService"), unit.category())

DATA: COUNT=col(source(s), name("COUNT"))

DATA: Churn=col(source(s), name("Churn"), unit.category())

COORD: rect(dim(1,2), transpose())

GUIDE: axis(dim(1), label("Customer's internet service provider (DSL, Fiber optic, No)"))

GUIDE: axis(dim(2), label("Percent"))

GUIDE: legend(aesthetic(aesthetic.color.interior), label("Whether the customer has left the ",
    "company (Yes/No)"))

GUIDE: text.title(label("Stacked bar chart of customer's internet service by churn status"))

SCALE: linear(dim(2), include(0))

SCALE: cat(aesthetic(aesthetic.color.interior), reverse(), include(
    "No", "Yes"))

ELEMENT: interval.stack(position(InternetService*COUNT), color.interior(Churn),
    shape.interior(shape.square))

```

END GPL.

SAVE OUTFILE='E:\Statistical Softwares\SPSS\CUSPC-SPSS\Assignments\Final
Project\Final_Project.sav'

/COMPRESSED.